MILITARY SPECIFICATION

TEST SET, INERTIAL MEASURING UNIT II, TS 3846/ASM-608(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification establishes the hardware and executive software design requirements for an Inertial Measuring Unit Test Set II (IMUTS II). The IMUTS II shall provide for maintenance support of the AN/ASN-92 Inertial Measuring Unit (IMU) and the AN/ASN-130 Inertial Navigation Group (ING) for both shore and sea based applications.

1.2 General The Inertial Measuring Unit Test Set II (IMUTS II), TS 3846/ASM-608(V), contractor's P/N 262200-1, shall consist of a two (2) bay console.

1.2.1 Test program sets (TPS) IMUTS II, when used with the appropriate Test Program Sets, shall provide for the test and fault isolation of those Weapons Replaceable Assemblies (WRAs) listed in table I. The TPS shall consist of those items necessary to test a Unit Under Test (UUT) on an IMUTS II. This includes the electrical, mechanical, instructional and logic decision elements. These individual elements and their function shall be as defined in the latest procurement contract for IMUTS II.

1.2.2 Control software. Control software for the Test Set shall be provided. Refer to paragraph 3.6.2 for a detailed description of the software requirements.

1.2.3 Self-check program. A Self-Check Program (SCP) and Self-Check Interface Device (ID) shall be provided.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents The following documents of the issue in effect on the date of invitation for bids, unless otherwise specified below, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (Code 9313), Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-N-18307	Aeronautical Electronic and Associated Equipment, Nomenclature and Nameplates for
MIL-N-81604	Navigation Set, Inertial (CAINS) AN/ASN-92(V)
MIL-T-28800	Test Equipment for Use With Electrical and Electronic Equipment, General Specification for

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers

Federal

FED-STD-595	Colors

Naval Air Systems Command

AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Litton G/CS

249467	Engineering Test Equipment, General Requirements
261896	Installation Requirements, IMUTS II

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Litton G/CS (Continued)

262283	IMUTS II ZIF Connector Input/Output Signal and Power Interface Characteristics
682021	ATS Programming Manual
------	FORTRAN Z Reference Manual
401492	Real Time Operating System User's Guide

(Application for copies should be addressed to Litton Systems, Inc., Guidance and Control Systems Division, 5500 Canoga Avenue, Woodland Hills, CA 91364.)

McDonnell Aircraft Company (MCAIR)

MDC A3818	F-18 Avionics Multiplex Design Specification

(Application for copies should be addressed to the McDonnell Douglass Corporation, P. O Box 516, Saint Louis, MO 63166.)

Electronic Industries Association (EIA)

RS-232-C-69	Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W , Washington, DC 20006.)

Institute of Electrical and Electronic Engineers (IEEE)

488-1975	Digital Interface for Programmable Instrumentation

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, 345 East 47th Street, New York, NY 10017)

2.3 Precedence of documents. When the requirements of the contract, purchase order, this document or referenced specifications herein are in conflict, the following precedence shall apply.

a Contract

b Purchase Order

c. Equipment Specification

d. Specifications referenced in a, b and c above.

All IMUTS II component part numbers referenced in this document shall be considered to be those in effect at the time of first article acceptance.

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3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish a sample unit for first article inspection and approval (see paragraph 4.4.2, First Article Demonstration Sample).

3.2 Parts, materials, and processes. Parts, materials and processes shall be selected to conform to the requirements of Litton Specification 249467 and the performance and reliability requirements of this specification.

3.2.1 Finishes. The IMUTS II front panels shall be finished as follows:

a. Contractor Federal STD 595, Color No. 26559 for door and panels, Color No. 26134 for side panels and trim.

b. Commercial units. These finishes shall conform to the requirements for contractor designed units or shall be in accordance with MIL-T-28800.

3.3 Design and construction. The IMUTS II shall be designed to best commercial practice in accordance with Specification 249467 In addition, contractor designed units shall be classified as Type III, Class 4, equipment in accordance with MIL-T-28800 with the following exceptions:

a. The enclosure for the overall test station shall be manufactured in accordance with approved engineering drawings.

b The ambient operating range shall be 10°C to 40°C

c Console air inlets may be less than 18 inches above the console mounting surface.

3.3.1 Physical characteristics. The IMUTS II subassemblies shall conform to the physical characteristics specified below

3.3.1.1 Total weight. The maximum weight for the IMUTS II shall not exceed 2000 pounds.

3 3.1.2 Dimensions/form factor. Each IMUTS II shall conform to the dimensions and form factor shown in the installation drawing 261896.

3 3.2 Connectors The IMUTS II shall utilize external connectors designed and selected to meet the requirements of MIL-T-28800 In addition, exposed male connector pins shall not carry voltages in excess of 30 volts when exposed.

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3.3 3 Electrical power sources and connections. The IMUTS II shall operate when supplied with 115V, 60 Hz and 115V, 400 Hz ac power meeting the requirements of MIL-T-28800, except that 400 Hz power shall meet the requirements of 400 ±20 Hz, 3 phase, WYE connected, ABC phase rotation, 5 wire (including Neutral and Earth Ground), 118 ±7V ac.

3.3 3 1 Power transient. The IMUTS II shall be capable of operating under the power transient conditions specified in MIL-T-28800 except that the duration of voltage and frequency transients shall not exceed 5 milliseconds. However, damage to IMUTS II requiring maintenance action shall not be incurred for durations specified in MIL-T-28800.

3.3.4 Modularity Placement of the IMUTS II modules shall provide for accessibility. Rear access shall not be required for the removal and/or replacement of its component modules or other major assemblies within the enclosure.

3.3.5 Internal controls and adjustments Internal controls of contractor designed units shall be easily accessible and shall be designed such that their position of normal adjustment is within the center 50 percent range whenever possible

3.3.6 Controls, displays and indicators Controls, displays and indicators shall be as defined by the contractor's engineering drawings.

3.3.7 Interference control. The IMUTS II shall be designed such that the requirements of MIL-STD-461 are considered to the extent that these requirements are consistent with best commercial design practice and the requirements of this equipment specification.

3.3 8 Reliability The Mean Time Between Failures (MTBF) of the IMUTS II shall be a minimum 300 hours for the contractor designed assemblies and 150 hours for the total station

3.3 8.1 Operating stability. The Equipment Verification Standard (EVS) shall be the reference source for calibration verification of the Digital Multimeter and the Frequency Counter/Timer. The Digital Multimeter and the Frequency Counter/Timer shall be subject to periodic calibration as determined by the EVS. The EVS is subject to calibration in accordance with paragraph 3.6 1.4 The Digital Multimeter and the Frequency Counter/Timer shall be used as a standard second source for all adjustments in the IMUTS II. The IMUTS II shall operate within specified performance for a period of 2000 hours or six (6) months whichever occurs first, continuously or intermittently, without the need for adjustment.

3.3.8.2 Total operating life. The IMUTS II shall have a minimum total operating life of 25,000 hours with reasonable servicing and replacement of parts.

3.3.9 Maintainability. The IMUTS II shall conform to the maintainability requirements of the maintenance plan.

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3.3.9.1 Standard test equipment. It shall be possible to locate faults in contractor designed units using only the test system itself and test equipments normally available in intermediate shop facilities.

3.3.9.2 Category of faults Failures in the IMUTS II shall be divided into four categories:

a. Category I· Faults which prevent program execution.

b Category II. Processor and peripheral faults which may be isolated with commercial supplied diagnostic programs.

c. Category III RTOS detected, in-process faults, which are out of scope for self-check.

d. Category IV Faults in the contractor designed assemblies, and commercial units which are controlled by contractor designed interface

3.3.9.3 Fault isolation Category I faults shall be isolated using applicable procedures and common ground support equipment existing in Intermediate (I) level maintenance shops.

Category II faults may be isolated to a functional area within the computer with the commercial supplied diagnostic programs.

Category III faults shall be automatically detected by the Real Time Operating System and alert the operator to the area and the nature of the fault.

Category IV faults are Self-Check Program (SCP) detectable and isolatible faults SCP shall detect 98% of all Category IV faults

When a Category IV fault is detected, the Self-Check Program shall isolate the fault to meet AR 10·

to 1 card, 90% of time

to 2 cards, 95% of time

to 3 cards, 100% of time

When cables, chassis mounted components, an instrument, etc., are potential probable causes of failure, they will be included in the component callout. Maintenance times for Category IV fault shall be consistent with MIL-T-28800.

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3.3.9.4 Adjustments. Adjustments are required when a device's parameters are out of tolerance but within the range of the adjustment to restore them to within tolerance. The need for adjustments is determined by a series of tests that are a subset of the Self-Check Program. Should a device's parameter be outside the range of adjustment available to restore that parameter to within tolerance, the device is considered to have failed. Devices subject to adjustment are:

a. Processor Unit power supply voltages.

b. Station fixed 5V dc supply voltages

c. Station multi-output fixed supply voltages.

d. Programmable power supply offset, gain, and output.

e Amplitude of 26V ac, 400 Hz reference.

f. Processor Unit A/D converter offset and gain.

3 3.9 5 Initial settings Certain devices have parameters that are set at time of installation and are not reset unless the device has been repaired or replaced These parameters are

a Actuation level of Processor Unit power supply ACLO signal

b. Over voltage and over current limits of all power supply outputs

c. Frequency and voltage tolerance limits of optional Power Monitor.

3 3.9.6 Power supply safety. Power supply failures in the IMUTS II shall not cause damage to any UUT or to the IMUTS II

3 3 9.7 Special tools The IMUTS II shall not require special maintenance tools except for cleaning and aligning the disc read/write heads

3.3.10 Service conditions. Console assemblies shall provide protection for subassemblies contained therein when subjected to the specified environmental conditions

3.3.10 1 Humidity Contractor designed units of the IMUTS II shall operate satisfactorily at the temperature and humidity requirements specified herein

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3.3.10.2 Vibration. The IMUTS II is intended for aircraft carrier, mobile van and shore based installation and shall meet the vibration requirements normally associated with transport of commercial equipment. With the requirements of MIL-T-28800 used as a design guide, the console shall meet the vertical vibration requirements. However, for horizontal vibration (fore-aft and side-to-side), the console may require additional support to limit the amplitude of sway.

3.3.11 Standard conditions. The following are the standard environmental conditions for demonstration of the IMUTS II.

a. Temperature: $25^{\circ}C \pm 5^{\circ}C$

b. Altitude: 0 to 4000 feet

c. Vibration None

d. Humidity: Up to 75% (without condensation)

e. Input Power: 118V ac ±7 volts, 400 Hz, 3 Phase
115V ac ±10%, 60 Hz

3.3.12 Interchangeability. The IMUTS II shall meet the interchangeability requirements specified in MIL-T-28800.

3.3.13 Safety. The IMUTS II shall be designed to meet the safety requirements of MIL-T-28800.

3.4 Nomenclature assignment. Nomenclature assignments for the IMUTS II shall be determined, when required, in accordance with MIL-N-18307

3.5 Functional description. The basic IMUTS II shall have the capability to test the Carrier Aircraft Inertial Navigation System (CAINS) Inertial Measurement Unit (IMU) WRA (P/N 680100-8 thru 20) of the AN/ASN-92 Inertial Navigation Set (INS) and the F-18 Inertial Navigation Group (ING) WRA (P/N 874500-2) AN/ASN-130 consisting of the Inertial Measurement Unit (IMU) (P/N 874210-3) and the Signal Data Converter (SDC) (P/N 874390-1). Litton G/CS shall furnish the Test Program Set (TPS) for the AN/ASN-92 Inertial Measurement Unit. McDonnell Aircraft Company (MCAIR) shall provide the software programs and interconnection devices (IDs) required to implement intermediate level automatic testing of the AN/ASN-130 using the IMUTS II

Test set operation and program execution shall be under control of the processor installed in the IMUTS II. The UUT application programs shall be stored on disc and executed in the processor. The test instructions shall be transmitted from the processor to the various modules in the IMUTS II via the processor/system bus network. Responses from test instructions shall be transmitted back to the processor via the same bus network.

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The processor shall execute interface control commands and select the Test Station Modules (TSMs) required to apply test stimuli and process UUT (Unit Under Test) response data. All TSMs shall interface with the processor via the processor/system bus.

The UUT shall be mounted on a fixture on the IMUTS II and tested via the associated interconnecting device (ID).

The operator control interface with the processor shall be provided by means of a CRT, keyboard, and printer.

3.5.1 Interfaces.

3.5.1.1 Electrical.

3.5.1.1.1 Primary power. The 60 Hz input power for the IMUTS II shall be provided via a 15 foot cable connected to the 60 Hz power switching unit. The three phase 400 Hz, input power shall be available by means of a 15 foot, five conductor cable from the 400 Hz switching unit.

3.5.1.1.2 Internal. The processor/system bus shall provide for the transfer of digital control and data signals between test equipment modules. In order to be processed by the processor unit, analog data shall be converted to digital format either within the test modules or via the digital multimeter or similar devices.

3.5.1.1.3 External. The electrical interface with the UUT shall be accomplished through a 1248 contact, zero insertion force (ZIF) connector. The IMUTS II ZIF connector shall be installed physically close to the UUT test fixture to minimize signal and power conductor length. The input and output characteristics at each ZIF contact shall be specified in the IMUTS II ZIF Input/Output Signal and Power Interface Characteristics, Litton Document No. 262283.

Each UUT shall interface at the ZIF connector by means of the associated interconnection device (ID). The ID shall provide the mating connectors, non-active devices (such as load resistors), and interconnect wiring required to conduct signals and power between the ZIF and the UUT.

3.5.1.2 Mechanical. Holding fixtures shall be affixed to the IMUTS II work surface to attach the UUTs in a manner similar to that specified in MIL-N-81604. The IMU test fixture shall provide a surface stable in all axes to $\pm$1.0 arc minute relative to the surface to which the test fixture is affixed. In addition, the IMU holding fixture shall be capable of leveling to 0 $\pm$5 degrees utilizing a bubble level visible without the IMU installed, to attain level within $\pm$1 degree. Each UUT ID shall be provided with mechanical locking devices relative to the work surface such that the device remains fixed in place after ZIF mating has occurred. The holding fixtures shall be designed to provide for UUT cooling air from the IMUTS II as required.

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3.5.2 UUT interconnection device (ID). Each ID shall contain the required UUT mating connector provisions, mounting surfaces, and test connector interfacing. Components in the ID shall be kept to a minimum. No active components shall be required within the ID.

3.5.3 Expansion capability. The basic IMUTS II shall contain sufficient physical volume and system wiring to provide expanded test capability with the addition of a third bay and/or another TSMB/ZIF connector combination.

3.6 Performance. Performance requirements for the IMUTS II subassemblies are specified herein. Table II lists the major hardware components by manufacturer and part number.

3.6.1 Test set hardware. The IMU Test Set II (IMUTS II) shall be a console containing the subassemblies listed in table II, or their equivalents.

3.6.1.1 Processor unit. The processor unit shall consist of a single chassis, processor power supply, module blocks, and the plug-in modules required to provide the following features:

3.6.1.1.1 Central Processing Unit (CPU).

CPU clock 8.43M Hz

Extended Instruction Set (EIS)
MULT/DIV/ASH/ASHC

Hardware floating point

Programmable real time clock

Mounting block

Memory management up to 128K words

3.6.1.1.2 Random Access Memory (RAM)

Capacity:	128K	
Typical cycle time:	500 ns	
Typical access time	Read	300 ns
	Write	200 ns

3.6.1.1.3 Diagnostic interface subassembly. The diagnostic interface subassembly shall interface the user with the processor driven test equipment. This interface provides the user with the capability to monitor the system performance, control the maintenance environment, and access the diagnostic programs that troubleshoot and isolate Category I, II, and III faults.

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3.6.1.1.4 Disc controller

3.6 1.1.4.1 Performance requirements The disc controller shall provide compatibility with system software and control the operation of the disc drive.

3.6.1.1.5 Output controller.

3 6 1.1 5 1 Performance requirements The output controller module shall provide the capability of transferring a 16-bit data word to one of eight external devices. The module shall provide a minimum of two input/output, read/write registers. The status register shall provide three device select bits (octal), eight (max) device level control bits, and three module control bits (start, reset, done). The data register shall contain up to sixteen bits of parallel data. The active high outputs shall be continuously enabled

3 6 1.1.6 General purpose interface bus (GPIB)

3 6 1.1 6 1 Performance requirements. The GPIB shall interface the processor bus with the digital multimeter (DMM) and counter/timer The GPIB module shall meet the requirements of IEEE Standard 488-1975 Logic levels shall be transistor-to-transistor-logic (TTL) compatible. Data transfer shall be initiated with an interrupt The interrupt level, 4 through 7, shall be selectable.

3.6.1 1 7 Bus buffer interface

3.6 1.1 7 1 Performance requirements The bus buffer shall interface the processor bus with the modules housed in the Test System Module Block (TSMB). The bus buffer unit shall consist of five sections

a Data transfer section The data transfer section shall contain the logic and control lines required to address and transfer data on the processor bus

b Interface section The interface section shall buffer the data and provide the control logic required to transfer data on the buffered bus network With respect to function and assertion level, the signals on the buffered bus shall correspond with those on the processor bus.

c Initialization section The initialization section shall generate reset signals to initialize all devices on the buffered bus network whenever the initializing command appears on the processor bus

d Priority arbitration section The priority arbitration section shall contain the control logic required to respond to bus interrupt, bus request, and bus grant signals on the processor bus.

e. Direct memory access (DMA) capability. The DMA section shall allow the insertion of the destination address back through the buffer to attain I/O control of the processor bus or the buffered bus in the TSMB.

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3.6.1.1.8 Ship's inertial navigation system interface.

3.6.1.1.8.1 Performance requirements The ship's inertial navigation system (SINS) transmits the ship's positional and velocity data via the ASW-25 data link. The SINS interface module shall contain the control logic required to process alignment data received over the data link. The interface shall reformat the data as required for retransmission over the processor bus.

3.6.1.1.9 Cathode ray tube interface.

3.6.1.1.9.1 Performance requirements. The cathode ray tube (CRT) interface shall be an asynchronous single-line interface capable of handling full or half duplex serial communication between the processor and the CRT/keyboard terminal. Switches shall be provided to facilitate selection of any one of the thirteen (13) standard data rates from 50 to 9600 Baud. The interface design shall conform to Electronics Industries Association (EIA) Standard RS-232-C. The CRT interface shall be capable of handling either local or remote (data only) communications for 8-level code devices.

3.6 1.1.10 Printer interface

3.6.1.1.10.1 Performance requirements. The printer interface shall be an asynchronous single-line interface capable of handling full or half duplex serial communication between the processor and the printer terminal. Switches shall be provided to facilitate selection of any one of the thirteen (13) standard data rates from 50 to 9600 Baud. The interface design shall conform to EIA Standard RS-232-C.

3.6.1.1 11 Analog to digital converter (ADC).

3.6.1.1.11.1 Performance requirement. The analog to digital converter (ADC) shall generate a 12 bit digital data word that represents the magnitude and sign of the analog input signals. The ADC shall provide for a minimum of 32 differential (dual) input signals. The magnitude (peak to peak) of the input signals shall be limited to ±10V. The resolution shall be 5.0 mV/bit.

3.6.1.2 Disc drive. A disc memory system shall be the mass storage medium for the IMUTS II.

3.6.1.2 1 Performance requirements. The disc storage capacity shall be a minimum of twenty million 16-bit words. The disc pack shall include 5 data surfaces and 1 servo surface and have an average access time of 30 milliseconds. The data transfer rate shall be 1 2 megabytes per second minimum.

3.6.1.3 Operator interface The IMUTS II operator interface terminal shall consist of a CRT display, input keyboard, and hard copy printer.

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3.6.1.3.1 Cathode ray tube/keyboard terminal.

3.6 1 3.1.1 Performance requirements.

On-Line Conversational Operation	Shall allow operator and computer to communicate directly (using the terminal as the connecting link)
Screen Capacity	1920 characters (24 lines of 80 characters per line)
Character Height·	.14 x .08 (standard 5 x 7 dot Matrix) inches
Number of Displayable Characters	67
I/O Asynchronous Interface	TTL RS-232 20/60 ma
Serial (Baud Rates)·	110-9600 (switch selectable)
Transmit Mode	10/11 BIT start-stop
Memory	Metal Oxide Semiconductor (MOS), 2048 character capacity
Display	12 inch cathode ray tube, TV type horizontal raster scan with P4 (white) phosphor
Cursor·	Blinking underline, alternates with displayed character at 4 Hz
Interface Code	USASCII 7-BIT
Hard Copy Output Connector	Located on rear panel
Remote Monitor Output Socket	BNC, located on rear panel

3.6.1.3.2 Printer terminal.

3.6.1 3.2 1 Performance requirements. The printer shall print hard copy test data and results on command from the processor. This printer shall be a non-impact, thermal, printing device capable of printing speeds up to 80 characters per second. A single print head shall be used to print one character at a time over an 80 column print line. Characters shall be formed by bringing the print head into light contact with heat sensitive paper and heating selected elements of the matrix.

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The printer shall also be equipped with BAUD rate switch and parity switch. The interface shall be serial TTL compatible. Paper capacity shall be 200 foot roll.

3.6.1.4 Equipment verification standard (EVS). The equipment verification standard shall be a precision voltage, frequency and resistance calibration source for on-site calibration verification of measuring instruments within the IMUTS II. The calibration standard shall be calibrated to the following accuracies and maintain this accuracy for one (1) year over a temperature range of 18°C to 28°C:

DC VOLTAGE

RANGE	ACCURACY		LOAD REGULATION	
	23°C±1°C 30 DAYS	23°C±5°C 1 YEAR	LOAD RESISTANCE	OUTPUT CHANGE (% OF SETTING)
μV (0-999 μV continuous; 0.2 μV resolution)	±2 μV	±2 μV		
1V (0.0 to 1.0 in 0.1V Steps) 0.1V - 0.9V	±10 ppm ±10 ppm	±20 ppm or 20 μV of range whichever is greater	100 MΩ 10 MΩ 1 MΩ	= 0.000% = 0.003% = 0.03%
10V (0 to 10V in 1V Steps) 1V - 9V	±7 ppm ±8 ppm			
100V (Cardinal Point)	±11 ppm		±5 ppm no load to full load	

TEMPERATURE COEFFICIENT		LINE REGULATION	SOURCE RESISTANCE*	RIPPLE
0°C to 18°C, 28°C to 50°C		±10% VOLTAGE CHANGE		
μV Range	±0.1 μV/°C	<1 μV	300Ω	<10 μV rms
1V Range	±5 ppm/°C	<1 ppm		<0.01% of range rms
10V Range	±5 ppm/°C	<1 ppm		
100V Range	±8 ppm/°C	<10 ppm	<1Ω (up to 0.5 mA load)	

*Output current will be function of source resistance, except 100V range which is limited at approximately 0.5 mA.
No damage to instrument with short circuit on output

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AC VOLTAGE

RANGE/FREQUENCY*	ACCURACY 30 DAYS 23°C±1°C	ACCURACY 1 YEAR 23°C±5°C	TOTAL HARMONIC DISTORTION & NOISE	OUTPUT CURRENT**
1V/400 Hz	±0.025%	±0.05%	<0.03%	0 to 10 mA rms
10V/400 Hz	±0.015%	±0.04%	<0.03%	
10V/4 kHz	±0.025%	±0.04%	<0 03%	
10V/50 kHz	±0.04%	±0.1%	<0.05%	
100V/400 Hz	±0.03%	±0.06%	<0.03%	0 to 0.5 mA rms

* Frequency ±1% except at 50 kHz ±5%.
** For load regulation stated below.

LOAD REGULATION		LINE REGULATION (±10% VOLTAGE CHANGE)	TEMPERATURE COEFFICIENT (0°C to 18°C, 28°C to 50°C)
1V Output (<1Ω source)	-0.005% (20 kΩ) -0.01% (10 kΩ)	<±10 ppm (All voltages at all frequencies)	<±25 ppm/°C (All voltages at all frequencies)
10V Output (0 to 10 mA)	±0.004% except ±0.008% @ 50 kHz		
100V Output (<30Ω source)	-0 006% (500 kΩ) -0 015% (200 kΩ)		

NOTE Current Limiting protects the 515A output from damage due to short circuit on the output.

RESISTANCE

RANGE (In decade steps)	ACCURACY 23°C±5°C for 1 year (Referred to "0" Ohms position)	TEMPERATURE COEFFICIENT 0°C to 18°C, 28°C to 50°C (Referred to Residual Resistance)	POWER RATING
0Ω	Residual Resistance <0 15Ω	<±0 4% /°C	0.2 watts or 100V dc or rms, whichever is less
10Ω - 100Ω	±0.06%	<±10 ppm/°C	
1 kΩ - 1 MΩ	±0.015%	<±5 ppm/°C	
10 MΩ	±0.075%	<±10 ppm/°C	

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OSCILLATOR OUTPUT

FREQUENCY	ACCURACY (10°C TO 40°C)	OUTPUT
1.00 mHz square wave	±10 ppm/year	TTL with 50 Ohm drive capability

3.6.1.5 Power monitor assembly. The power monitor panel shall contain monitoring capability for both 60 Hz and 400 Hz primary power. This monitor shall continuously and visually display voltage and frequency of primary power and shall detect and visually display out-of-tolerance conditions. Software interfaces shall also be provided to generate appropriate messages on the CRT and printer to indicate that a particular voltage or frequency has been out of tolerance (high or low) for at least 5 ms.

3.6.1.6 Programmable UUT power supply. The requirements for the programmable power supply include the following:

a. Input 115V ac ±10%, Single Phase, 47-66 Hz.

b. Output.

#1 voltage range 0 to +40V dc, 15 amps, 600 watts max.

#2 voltage range 0 to -40V dc, 5 amps, 200 watts max.

c. Output voltage shall rise linearly from 10 percent to 90 percent level within 10 milliseconds

d. Temperature coefficient: 0.03%/°C max (0.005%/°C typical).

e. Line regulation 0.02 percent + 2mV over primary voltage input range of 115V ac ±10 percent.

f. Load regulation 0.02 percent + 2mV no load to full load

g. Ripple shall be no greater than 0.1% rms

h. Remote programming shall be capable of precisely selecting any output voltage from 0 to full scale in less than one second. The scale factor error and linearity shall both be within 0.5% of nominal. The resolution required to achieve these accuracies shall be available with a maximum of 12-bits of binary data. Optical isolation shall permit the grounding of either power supply output without concern for the grounding of the programmer unit's digital inputs by using -sense and -sense terminals.

j. Overvoltage protection shall be provided

k. External fusing of the input, at twice the rated output power, shall be provided on all power supplies without self-contained protection.

l. External overload protection shall be provided on all regulated outputs by electronic current limiting with automatic recovery.

m. Thermal protection shall be provided.

n. Voltage and current meters and overvoltage lights shall be provided for each output.

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3.6.1.7 Set power supplies. The following voltages and current capacity shall be supplied to furnish power to the test system modules:

a. Input to power supply: 115V ac ±10V, single phase, 47-63 Hz

b. Output:

+5V dc at 100 amps

+15V dc at 4 amps
-15V dc at 4 amps

+35V dc at 6 amps
-35V dc at 6 amps

+10V dc at 1 amp
-10V dc at 1 amp

+20V dc at 1 amp
-20V dc at 1 amp

+28V dc at 10 amps

The following requirements apply to each output

c. Voltage adjustment over a range of ±15% shall be provided.

d Temperature coefficient shall be 0.03%/°C maximum. Typical value shall be 0.005%/°C

e Line regulation shall be 0 1% over a primary line voltage input variation of 115V ac ±10%.

f Load regulation shall be 0 1% from no-load to full load.

g Ripple shall not be greater than 0 02% rms

h Remote sensing shall be provided.

i. Overvoltage protection shall be provided.

j Overload protection shall be provided by electronic current limiting with automatic recovery

k Thermal protection shall be provided.

3.6.1.8 Test set module block. The TSMB shall provide for UUT stimuli generation and for UUT response monitoring. The unit shall contain the UUT interconnecting device Zero Insertion Force (ZIF) interface connector. Signal inputs to and outputs from external TSMs shall be routed through the TSMB. The TSMB cardcage shall be designed to facilitate front access maintenance of the TSMs and chassis mounted components.

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3.6.1.8.1 TSMB configuration. The IMUTS II TSMB shall contain plug-in modules (TSMs) that provide the signal interface functions between the buffered bus network and the UUT. The internal TSMs housed in the TSMB shall conform to the standard hex card form with integrated circuits mounted on PC board sockets, except for the Relay II TSM, whose components shall be soldered to the board The TSMB backplane shall provide a common interconnect with the buffered bus network and module power and ground systems.

3.6.1.8.2 TSMB performance requirements. The minimum performance requirements for the modules of the TSMB include the following specifications:

a. Analog Multiplexer (MUX). The analog MUX shall provide the capability to select automatically signals from either the ZIF connector or internal test points and present these signals to an instrument for measurement or further processing.

b Termination Network The termination network shall provide a selection of passive devices (load resistances, R/C filters, diodes, etc.) which may be required to load or condition analog data for measurement by IMUTS II The selection of the individual termination device and the required measuring instrument shall be under program control

c. Synchro/Resolver Input. The synchro/resolver input module(s) shall provide for the conversion of synchro or resolver data into 16-bit (minimum) digital format. Under program control, one of ten channels of synchro, or resolver data may be selected for conversion The selection of reference voltages shall be under the control of the software. Eight channels shall provide 16-bit resolution with the Most Significant Bit (MSB) representing 180° and the Least Significant Bit (LSB) representing 0 0055°. A high speed channel shall provide 18 bits of resolution with the LSB weighted at 0.0014° The tenth channel shall be a high speed converter with a resolution of 20 bits (LSB = 0.0003 deg).

d Analog Multiplexer (AMUX) Bus. The AMUX bus module(s) shall provide two AMUX communications channels which conform to the requirements of MDC-A3818 The AMUX bus module(s) shall be capable of sending and receiving a maximum of 32 individual message formats. The AMUX bus module(s) shall provide circuitry to detect AMUX bus faults within the UUT. The AMUX modules shall also stimulate common AMUX faults as defined in MDC-A3818.

e. SDC Interface. The TSMB shall contain module(s) capable of communication with the Signal Data Converter (SDC) bus in the AN/ASN-130. Module design shall provide for DMA and programmed I/O data transfers. The critical timing relationships and limited drive capability of the SDC bus shall be considered in the design of the SDC interface module(s)

f. Power Control. Module(s) within the TSMB shall monitor and control the application of primary power to the UUTs. Out of tolerance UUT voltage and/or frequency conditions shall result in an automatic, safe shutdown of UUT power. Selected no-go discretes from the UUT or test station shall

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also result in an automatic shutdown of power to the affected unit. The faulty power or no-go condition shall be stored in a status register. The processor shall be alerted to the presence of a fault by the automatic generation of an interrupt on the buffered bus

g. Internal Measuring Unit (IMU) Memory Interface Module(s) shall be provided in the TSMB which are capable of performing a direct memory access (DMA) read/write, 22-bit serial data transfer with the AN/ASN-130 IMU memory. The module(s) shall read and write the IMU calibration constants and verify the IMU memory performance.

h. Delta Function Counter. The delta function counter shall be capable of simultaneously counting up to three linear velocity and three angular velocity signals generated by the UUT. Control over the counting interval shall reside in software The ΔV counters shall be capable of counting up to a minimum of ± 2047 pulses in each time interval. The $\Delta\omega$ counter shall be capable of counting at least ± 127 counts before requiring a reset.

i. Torquer/Slew Simulation Module(s) in the TSMB shall simulate gyro torque and gimbal slew pulse inputs to the CAINS IMU The length of the differential pulses shall be proportional to the magnitude of the binary word loaded by the program.

j Discrete Input/Output Module(s) shall provide 64 bi-state receiver/driver channels, eight Double Pole Double Throw (DPDT) relay outputs, and six tri-state outputs Under program control, the bi-state receiver/drivers shall be capable of either sending or receiving a 32 channel broadside of discrete signals. The discrete signal amplitudes shall have a range of 0 to +30V dc

k. No-Go Monitor. Module(s) within the TSMB shall provide for the automatic monitoring of UUT no-go signal discretes Through the power control function, selected no-go conditions shall initiate an automatic UUT power shutdown sequence The no-go monitor shall have the capability of storing no-go status, generating interrupts on the detection of a no-go signal, and masking or inhibiting selected no-go conditions.

3.6.1.9 Frequency counter-timer The programmable frequency counter-timer shall perform precise measurements of frequencies and periods for selected test signals The counter-timer shall also have the capability of precise measurement of the time interval between discrete events Full remote programming capability shall be provided through the IEEE STD 488 GPIB interface. Dual input channels shall be provided with the ability to select separate or common inputs between channels The trigger-level controls for channels A and B shall provide for the selection of the positive or negative slope trigger level. A parallel data output option shall be provided for digit, decimal point, overflow, and units annunciation information for data acquisition. The following specifications apply at the counter/timer terminals:

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3.6.1.9.1 Frequency measurements.

a.	Range:	0 to 100 MHz minimum
b.	Gate Time:	1µs to 10s in 6 decade steps
c.	Resolution:	0.1 Hz at 10s gate time to 10 kHz at 0.1 ms
d.	Accuracy:	Timebase accuracy ±1 count

3.6.1.9.2 Ratio measurements.

a.	Display:	f_1/f_2, where f_1 and f_2 are applied to inputs channels A and B respectively
b.	Range.	f_1 (0 to 100 MHz), f_2 (0 to 100 MHz)
c.	Accuracy	±1 count of signal on ch. A ± trigger error of signal on input B

3.6.1.9.3 Period measurements.

a.	Range:	10ns to 10^9s
b	Periods Averaged:	1 period to 10^8 periods
c.	Freq. Counted:	1 Hz to 10 MHz
d	Resolution:	10ns
e.	Accuracy:	±Timebase accuracy ±1 count ± trigger error

3.6.1.9.4 Time interval measurement.

a	Range	10ns to 10^9s
b.	Input:	Channels A and B, common or separate
c.	Resolution:	10ns
d	Accuracy:	±1 count ± timebase accuracy ± trigger error

3.6.1.9.5 Totalize measurement.

a.	Frequency Range:	0 to 100 MHz
b.	Count Range:	0 to 10^9
c.	Accuracy:	±1 count per gate

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3.6.1.9.6 Input characteristics.

Channels A and B

a Sinewave Sensitivity. 25 mV rms from dc to 1 MHz
50 mV rms to 50 MHz
100 mV rms to 100 MHz

b Pulse 150 mV pulse amplitude, with minimum pulse width of 8 ns

c. Input Impedance 1 MΩ in parallel with 25 pF max

d Timebase 10.0 MHz

e Voltage Ranges: autoranging

3.6.1.9.7 Rear inputs The unit shall provide rear inputs for Channels A and B.

3.6.1.9.8 Output characteristics.

a Display. The result of any measurement shall be shown on an 8-digit, minimum, visual display

b Digital output unit The counter-timer shall provide GPIB compatible 8-bit parallel outputs from each digit, plus overflow, units annunciation, decimal point, and print command.

3.6.1.9.9 Remote programming GPIB programming of range, mode, slope, and reset functions shall be provided along with analog programming of trigger levels, power sense, overflow status, and system ready outputs The ac/dc coupling, attenuation, and separate or common mode shall be programmable. The trigger level for Channels A and B shall be programmable over a +1V to -1V range with a resolution of 1% and an accuracy of 5% plus 2 mV Two analog input/output lines shall be provided to check D/A performance

3.6.1.10 Digital multimeter (DMM). The programmable digital multimeter shall perform precise measurements of DC volts, AC volts, and resistance. The IMUTS II DMM shall meet the following minimum performance requirements for each measurement task at the DMM input terminals

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3.6.1.10.1 DC voltage.

ACCURACY (Normal Resolution) ±(% of Input Plus Number of Digits)

Range	Full Scale	Resolution	Accuracy		
			24 Hours (23°C ±1°C)	90 Days (18°C to 28°C)	1 Year (18°C to 28°C)
100mV	312 mV	1 uV	0.002 + 4	0 003 + 5	0.005 + 8
1V	2.5V	10 uV	0.001 + 1	0.002 + 1	0.004 + 1
10V	20V	100 uV	0.001 or 1	0.001 + 1	0.002 + 1
100V	160V	1 mV	0.001 + 1	0.002 + 1	0.004 + 1
1000V	1200V	10 mV	0.001 + 1	0.002 + 1	0.004 + 1

3.6.1.10.2 AC voltage. This parameter shall be measured as Root Mean Square (RMS) value for sine waves and average value for other waves Minimum performance requirements are as follows

ACCURACY (±(% of Input + No. of Digits)

Frequency	Accuracy (0 1% of Range to Full Scale)					
	24 Hours (23°C±1°C)		90 Days (18°C to 28°C)		1 Year (18°C to 28°C)	
	0 to 500V	Above 500V	0 to 500V	Above 500V	0 to 500V	Above 500V
30 Hz to 50 Hz	0.3 + 5	0.4 + 5	0.5 + 5	0.55 + 5	1.0 + 5	1.0 + 5
50 Hz to 10 kHz	0.035 + 5	0.075 + 5	0.05 + 5	0.1 + 5	0.1 + 5	0.2 + 5
10 kHz to 40 kHz	-	0.1 + 5	-	0.15 + 5	-	0.3 + 5
10 kHz to 50 kHz	0.075 + 5	-	0.1 + 5	-	0.2 + 5	-
50 kHz to 100 kHz	0.3 + 5	-	0.5 + 5	-	1 0 + 5	-

INPUT CHARACTERISTICS

Range	Full Scale	Resolution		Impedance
		Normal	High	
1V	2 5V	10 uV	1 uV	1 MΩ <100 pF
10V	20V	100 uV	10 uV	
100V	160V	1 mV	100 uV	
1000V	1000V	10 mV	1 mV	

3.6 1.10.3 Resistance. Capability for four terminal measurements shall be made on eight ranges from 10 ohms to 100 megaohms. Accuracies up to ±0.003% +1 digit and resolutions up to 100 micro ohms shall be obtainable as follows

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Resolution			Accuracy ±(% of Input) 1 Year (18°C to 28°C)
Range	Normal	High	
10Ω	100 uΩ	-	0.01% + 20 digits
100Ω	1 mΩ	100 uΩ	0.006% + 2 digits
1 kΩ	10 mΩ	1 mΩ	0.006% + 1 digit
10 kΩ	100 mΩ	10 mΩ	0.006% + 1 digit
100 kΩ	1Ω	100 mΩ	0.006% + 1 digit
1 MΩ	10Ω	1Ω	0 006% + 1 digit
10 MΩ	100Ω	10Ω	0.04% + 1 digit
100 MΩ	1 kΩ	100Ω	0.1% + 1 digit

3.6.1.11 Primary power. The primary power requirements for the IMUTS II shall be·

a. Source No. 1

Voltage	118 ±7V rms line-to-neutral
Frequency	400 ±20 Hz
Phase	3 Phase, 5 wire, Wye connected
Current	10 Amps/phase
Power Factor	Unity

b. Source No. 2

Voltage	115V rms ±10 percent, line-to-neutral
Frequency	60 ±3 Hz
Phase	Single
Current	50 Amps
Power Factor·	0.75 minimum

3.6.2 Software description.

3 6.2 1 System software. The system software for the IMUTS II shall consist of the following modules

a. Real-Time Operating System (RTOS)

b. Execution Executive

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c. Fortran Z Compiler

d. File Manager

e. Source Editor

f. IMUTS Library

g System Library

h. ATLAS Compiler

3.6.2.1.1 Real-time operating system (RTOS). The real-time operating system shall be the RTOS dual word system. The RTOS controls the system and responds to inputs from the keyboard. Communication shall be via the CRT/keyboard terminal. Via this interface, the operator shall inform the system which test program is to be utilized or shall initiate execution of any option in the system. The RTOS shall provide resource allocation, job control and file maintenance for the test programs. It shall be an interface between the test program, the test station, and the operator. A status display showing resource utilization data shall also be provided.

a. Resource allocation IMUTS resources shall be grouped as follows:

CPU time
Main memory
System hardware

Resources shall be allocated to user programs on a first-come, first served basis within a priority system. There shall be eight levels of priority available at the CPU level, assigned as follows:

7 unassigned (highest priority)
6 real-time clock
5 disc memory
4 all other hardware
3 unassigned
2 reserved
1 CPU
0 all user tasks

b. CPU time. CPU time shall be allocated to competing tasks in such a manner that it appears to the user that several tasks are being executed simultaneously. Because it is unlikely that any one given task can utilize the full processing capability of the CPU, RTOS shall provide the means of deactivating a task which is not using the CPU in favor of activating another task that is ready to use the CPU, even though the second task may be of lower priority than the first.

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Tasks that have been temporarily deactivated for not using the CPU shall reclaim their priority when ready to resume execution either by enabling an external device to interrupt the lower priority tasks, or by enabling the internal real-time clock to signal the passage of a specified time interval.

c. Main memory The main memory shall be distinguished from other forms of storage in that a program must reside in main memory to be executed. All UUT programs shall be allowed use of a common area of main memory as a working space When the UUT program references a subroutine that is not resident in main memory, that program shall be deactivated in favor of other programs waiting for CPU time until the referenced subroutine is retrieved from the disc

d. Status display The top two lines of the IMUTS II CRT screen shall be used to display regarding CPU time and main memory utilization in accordance with document 401492

e. Job control The job control shall provide the user with the capability of selecting the operating program desired for the IMUTS II. It shall be the computer software interface by which the user may select and execute one of the program options to manipulate files, or to manipulate or initiate the execution of a UUT test program The job control shall allow the user to run programs in real time

f. File maintenance RTOS shall provide the means by which the operator may create or delete entries in a directory of programs catalogued on the disc.

g Bootstrap A bootstrap shall be provided to start the station and to initialize RTOS The bootstrap shall be loaded automatically via a Read Only Memory (ROM). When the bootstrap is executed it shall load the absolute loader which resides in low core The absolute loader shall then initialize the system

h Atlas compiler It shall permit the user to compile FORTRAN

3.6 2 1 2 Execution executive The execution executive, known as the COMMA BGATR (Begin Automatic Test Routine) shall provide test sequence control, test program status display, operating modes, and operator-initiated printer annotation Test sequence control elements shall be the executive program, the test program, a control file, and operator input commands. The control file for each test program shall define a series of subroutines or portions of subroutines that constitute the entire test program Each entry in the control file shall define a point of possible action by the executive program, which is initiated by and runs concurrently with the test program.

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a. Status display. Test program status display shall be provided continuously by the executive program on the CRT display. Displays shall include the selected mode of operation, the elapsed time since the beginning of the current test and the results of the last test Measured or computed parameters associated with the unit under test shall also be displayed. Printer annotation shall allow the operator to record failed test results and comments on the hard copy printout.

b. Operator interfaces. The operator interface shall be the CRT/keyboard terminal. All CRT displays shall be self explanatory to the operator.

3.6.2.1.3 FORTRAN Z compiler. The FORTRAN Z compiler, P/N 862514 shall be included in the IMUTS II disk. The FORTRAN compiler shall be installed on the station to allow source-to-object compilation. The output of the compiler shall be machine code. The source language shall be a subset of the ANSI FORTRAN IV as described in the Litton FORTRAN Z reference manual. The FORTRAN compiler shall produce source listings with compiler assigned statement numbers.

3.6.2.1.4 File manager. The "MAINT" subsystem in RTOS shall serve as the file manager and shall provide the following capabilities:

a. List the current directory.

b. Add files to the disk.

c. Delete files from the disk

3.6.2.1.5 Source editor. A source editor shall be provided to create and edit source files on-line. Editor features shall be as follows

a. Line editing:

(1) Line/multiple line deletion

(2) Line replacement

(3) Line insertion

b. Character editing (characters and character strings)

(1) Character deletion

(2) Character replacement

(3) Character insertion

c. File merging

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Source editor operation shall be based on the CRT display/keyboard, using the CRT to display the pending line being edited, and the keyboard as the operator command input device All editing shall be done on a scratch file, with commands provided to create a scratch file from an existing file and to store the edited scratch file under either a newly created file name or an existing file name, replacing the previous file contents with the contents of the scratch file.

3.6.2 1 6 IMUTS library. The IMUTS library shall be P/N 260865, as a baseline, and updated to include any required changes for IMUTS II. This library shall contain general purpose subroutines used by the IMU application test program

3.6.2 1.7 System library. The system library shall be P/N 260864 as a baseline and updated to include any required changes for IMUTS II.

a. RTOS FORTRAN library This shall be a collection of mathematical subroutines which support the solution of complex calculations It shall include the FORTRAN format scanner which shall be used to translate format statements into the specific appropriate action required The library shall be resident on the disc.

b RTOS run-time library This shall be a collection of subroutines which include the general purpose input/output subroutines and the test station equipment drivers. Input/output routines shall include those required to write on the CRT display and the serial printer and read the keyboard. Test station equipment drivers shall include subroutines for selecting the analog scanner channel, programming and reading the DMM, and other test station equipment drivers

3 6.2.1 8 ATLAS preprocessor. The ATLAS preprocessor shall be used to translate ATLAS source statements defined in Litton document 682021 into code consisting of FORTRAN statements

The preprocessor shall be written in FORTRAN language, thus allowing maximum portability for both the preprocessor and its output code

3.6.2.1.8.1 Operation. The ATLAS preprocessor shall perform a textual scan of each source statement to identify variable names and labels. These shall be placed in the symbol table in lexicographical (alphanumeric) order. Duplicate and illegal symbols shall be flagged by an error message A syntax analysis shall then be performed to identify the statement type and to parse it into its elementary components

The syntax analyzer shall be driven by a table consisting of syntax specifications. Illegal or unrecognized syntax shall be flagged by an error message. The FORTRAN intermediate code shall then be generated by a table look-up procedure. The code thus produced shall be scanned from the start in order to supply the hard

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copy listings and to complete the code generated by forward branching. Undefined or doubly defined branch points shall be flagged with an error message.

Source code and errors shall be generated in the form of FORTRAN comments in order to avoid interfering with the operation of the FORTRAN compiler, which will be processing the ATLAS output code

3.6.2.2 Applications programs.

3.6.2.2.1 IMUTS II self-check program. A self-check program for IMUTS II shall be developed that provides for fault detection and isolation of all Category IV faults to a replaceable assembly or item of commercial equipment. For TSMs, the design goal of the self-check program shall be to isolate to the level required under AR-10A.

3.6.2.3 IMUTS II operating modes. The IMUTS II shall provide a flexible method of testing a UUT. The station shall provide to the operator a choice of several operating modes and options within these modes

3.6.2.3.1 Operating modes. The operating modes are:

a. AUTO. In this mode the test program will execute end-to-end without operator intervention unless requested by the test program software. This is the normal mode in which to execute a Ready-for-Issue test.

b. SEMI: In this mode the test station will execute one or more tests as determined by STEP size (refer to 3.6.2.3.2b), and then pause for operator action.

c CAM This is the Computer Aided Manual mode It provides the maximum amount of interaction between the operator and the test station The operator has control of test station hardware in this mode.

3.6.2.3.2 Options.

a. FAIL STOP: This option is associated with AUTO mode. If selected, the program will execute as in AUTO mode unless a failure is encountered, at which time the program stops and waits for operator action. If FAIL STOP is not selected, the program executes to a conclusion (failure or RFI), the appropriate data is printed, and the CRT display is extinguished.

b. STEP. This option allows the operator to select the size of the test when in SEMI or CAM modes The size may be a single test (TEST), a set of tests (SET), a group of sets (GROUP), or the whole program. The default state is the program.

c. REPEAT: The operator has the option to select the number of times (1 to 99) STEP is to be repeated. The default state is one time.

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3.6.2.4 Printer operation. The normal operation of the printer shall be to print only that which is imbedded in the application code as calls to the printer. However, the results of each test measurement are temporarily stored on a disc file and can be caused to be printed by the command EXEC RETR, which is an acronym for EXECUTIVE RETRIEVE. The operator may type EXEC RETR on the keyboard at the end of a successful RFI program execution in AUTO mode, or when a failure occurs in any mode, or at the end of execution of a STEP size program in SEMI or CAM modes. Should the operator elect to begin another program without retrieving the data from the previous program, the previous program's data are written over by data from the current program.

3.7 Documentation.

3.7.1 Engineering drawings. A full design disclosure set of engineering drawings shall be developed to the requirements of MIL-D-1000, Form 3. The following types of engineering drawings shall be developed and maintained under configuration management in accordance with MIL-STD-480.

a. Detail fabrication drawings

b. Assembly drawings

c Wire lists

d. Schematics (Form 4)

e. Software control drawing(s)

f. Program listings

3.7.2 Equipment test procedure (ETP) The ETP shall be developed by the contractor. The ETP shall specify all the prerequisites which must be satisfied prior to the beginning of the test. It shall include the step-by-step procedure which, when successfully performed, will ensure compliance of the IMUTS II with this specification The ETP shall contain data sheets for entry of quantitative test data and provisions for indicating pass or fail for each test observation.

3.7.3 Calibration procedure Items requiring periodic calibration per Calibration Maintenance Requirements Summary (CMRS) are

a Digital Multimeter

b Counter/Timer

c. Equipment Verification Standard

For further clarification, see paragraph 3.7.4.

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3.7.4 Equipment verification procedure. A verification procedure shall be prepared by the contractor which delineates the verification requirements of the IMUTS II. This procedure shall involve use of the Equipment Verification Standard (EVS) and is valid only for IMUTS II stations so equipped. Items whose status will be verified by the EVS are:

a. Digital Multimeter

b. Counter/Timer

Should the measurement accuracies of the above devices be found by the EVS to be sufficient for UUT and Self-Check Program (SCP) measurements, periodic calibration of the above devices, as described in paragraph 3.7.3, shall not be necessary.

3.8 Workmanship. Workmanship shall be in accordance with generally acceptable commercial criteria for this category of equipment. There shall be no imperfections that could reasonably be expected to prevent the requirements of this specification from being met. All fittings, assembly processes, coatings, and finishes shall meet the workmanship criteria of the applicable supplier. There shall be no evidence of detrimental sharp corners or edges, burrs, dents, scratches, gouges, cuts, cracks, or other similar conditions Fastening, clamping, supporting, and locking devices shall be correctly installed without damage. There shall be no detrimental foreign matter internally or externally. Markings shall be non-injurious, legible, and of necessary permanence.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government.

4.2 Classification of tests Inspection of the IMUTS II shall be classified as follows:

a. Individual components. Individual component inspection consists of examinations and tests performed on each component of the IMUTS II submitted for acceptance.

b. First article inspection. First article inspection consists of examination and tests performed after award of the contract on a sample IMUTS II to determine that the set meets the requirements of this specification.

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c. Quality conformance inspection. Quality conformance inspection consists of examination and tests performed on each IMUTS II submitted for acceptance under contract

4.3 Inspection conditions. Unless otherwise specified, the following are the standard test conditions for demonstration of the IMUTS II.

a	Temperature	$25^{\circ}C \pm 5^{\circ}C$
b	Altitude	0 to 4000 feet
c	Vibration	None
d	Humidity	Up to 75% (without condensation)
e	Input Power	115 Vac ± 2, 60 Hz, 1 phase 115 Vac ± 7 volts, 400 Hz, 3 phase

4.4 General test procedures

4.4.1 Individual components. Individual components, as specified herein, will be accepted separately and independently. Each component of the IMUTS II shall be inspected to assure compliance with the issue of engineering drawings in effect at the time of inspection

4.4.2 First article demonstration sample. The first article demonstration sample shall consist of one complete IMUTS II test set. The sample shall be identical in design and performance to the items which will be produced on the manufacturer's production line. The acceptance test for the first article sample shall consist of successful demonstration and completion of the Equipment Test Procedure (ETP)

4.4.3 Quality conformance inspection. The quality conformance inspection for production IMUTS II units shall be in accordance with the contractor's quality assurance plan

4.5 Cause for rejection. Failure of the set to meet any of the requirements specified herein shall be cause for rejection.

4.6 Rejection and retest. Sets which have been rejected may be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejecting and the action taken to correct the defects found in the original test shall be furnished to the inspector

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5. PACKAGING

5.1 General. When prepared for shipment, all major components and parts of the IMUTS II shall be preserved, packaged, packed, and marked for shipment in accordance with the requirements of MIL-E-17555, Level C.

6. NOTES

6.1 Intended use. The IMUTS II covered by this specification is intended for use by Naval and Marine personnel in the performance of testing for maintenance support of the AN/ASN-92 Inertial Navigation Set and AN/ASN-130 Inertial Navigation Set at both shore and sea-based facilities. The IMUTS II is primarily designed for use under the environmental conditions specified herein.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. First article inspection sample (4.4.2).

c. First article inspection sample size (4.4.2).

d. First article inspection data or test reports.

6.3 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.7.1	Engineering drawings	DI-E-23174
3.7.2	Equipment test procedure	DI-T-2144
3.7.3	Calibration procedure	UDI-T-20401
3.7.4	Equipment verification procedure	

6.4 First article When a first article is required, it shall be tested and approved under the appropriate provision of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample consisting of one IMUTS II unit as described herein. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article.

6.5 Abbreviations and acronyms.

ACK	Acknowledge
ADC	Analog to Digital Converter
ADRS	Address

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ASH	Shift Arithmetically Instruction
ASHC	Shift Arithmetically Combined Instruction
ANCU	Aircraft Navigation Computer Unit
BGATR	Begin Automatic Test Routine
BPI	Bits Per Inch
CAINS	Carrier Aircraft Inertial Navigation System
CAM	Computer Assisted Manual
CMRS	Calibration Maintenance Requirement Summary
CPU	Central Processor Unit
CRC	Cyclic Redundancy Check
CRT	Cathode Ray Tube
CYC	Cycle
DMA	Direct Memory Access
DMM	Digital Multimeter
DPDT	Double Pole Double Throw
ECC	Error Correction Code
EVS	Equipment Verification Standard
FAD	First Article Demonstration
GSERD	Ground Support Equipment Recommendation Data
GPIB	General Purpose Interface Bus
ID	Interconnection Device
IMU	Inertial Measuring Unit
IMUTS	Inertial Measuring Unit Test Set
I/O	Input/Output

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IPS	Inches Per Second
KB	Keyboard
KHZ	Kilo Hertz
LSB	Least Significant Bit
MHZ	Mega Hertz
MOS	Metal Oxide Semiconductor
MSB	Most Significant Bit
MTBF	Mean Time Between Failure
MUX	Multiplexer
NPR	Non-Processor Request
NSTS	Navigation Set Test Station
PCOF	Probable Cause of Failure
ROM	Read Only Memory
RTOS	Real Time Operating System
SCP	Self Check Program
SDC	Signal Data Converter
SINS	Ship's Inertial Navigation System
SPC	Small Peripheral Controller
SRA	Shop Replaceable Assembly (AR-10)
TPI	Test Program Instructions
TSM	Test Station Module
TSMB	Test Station Module Block
TTL	Transistor-to-Transistor Logic
UUT	Unit Under Test
WRA	Weapons Replaceable Assembly (AR-10)
ZIF	Zero Insertion Force

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TABLE I. WRA Unit Under Test

Litton Part Number	Unit
680100-8 thru -20	CAINS IMU (CN-1263/ASN-92(V))
874500-2	F-18 ING (AN/ASN-130)

TABLE II. IMUTS II Major Components

Unit	Manufacturer	Manufacturer Part Number	Litton Part Number
Equipment Verification Standard	Fluke	515A	924420
CRT Display and Keyboard	Tec, Inc	0455-22204Y 0711000007/09557	989170-5
Printer	Electronic Marketing Specialists	TO-83	989179-1
Processor Unit	Litton		262208-1
Disc Drive System	Plessey Peripheral	701835-102	923612-20
Test Station Module Block (TSMB)	Litton		262210-1
Power Monitor Assy	Litton		262290-1
0 ±40V dc Prog Power Supply	Asto-Geo-Marine	PS 361	924468-1
Timer/Counter	Dana Lab	9015/11A/S-4475/01-55/60	923392-3
Digital Multimeter	Fluke	Model 8500-A/AE-01/02/07/08	253577-1
Station Power Supply (Mult)	Astro-Geo-Marine	PSF 363	924468-4
Station Power Supply +5V dc	Astro-Geo-Marine	PSR 362-1	924468-2
Power Switching (400 Hz)	Litton		261548-2
Power Switching (60 Hz)	Litton		261548-1
Component Assy, Power	Litton		262285-1
Diagnostic Interface Sub-Assembly	Dynamic Sciences Int.	204-1049	989178-1

Preparing Activity:
NAVY-AS
(Project No. 6625-N790)

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APPENDIX

INPUT/OUTPUT SIGNAL CHARACTERISTICS FOR THE INERTIAL MEASURING UNIT TEST SET II

10. SCOPE

The IMUTS II Automatic Test Set incorporates a set of interface modules that provide the signal input and output (I/O) capabilities at the zero insertion force (ZIF) connector. This appendix specifies the signal characteristics at the ZIF interface.

20. GENERAL

20.1 Test Station Outputs

Figure 1 is a simplified block diagram of the IMUTS II in the standard test configuration. During the automatic test mode, test sequences are initiated within the Processor Unit The Processor Unit sends control signals to the peripheral devices and over the buffered bus network. Transceivers in the individually selected Test Station Modules (TSMs) receive control and test data signals and transmit Unit Under Test (UUT) response data over the buffered bus. The TSMs, which are housed in the Test Set Module Block (TSMB), define the test station input/output characteristics at the ZIF The UUT receives the test station outputs via its Interconnect Device (ID). The ID provides the hardwired interconnections between the UUT and the ZIF

20.2 Test Station Inputs

Response data from the UUT enters the test set at the ZIF via the ID The Processor Unit directs selected TSMs to receive and process the data Depending on test requirements, the data may be measured, counted, compared, stored, and, if analog, converted to a digital format prior to being transmitted to the Processor Unit.

20.3 Input/Output Characteristics

Table I describes the performance characteristics at the test sets' ZIF contacts (refer to directional arrows in Figure 1). The data listed under the heading "RANGE" are typical of the input/output device under nominal load conditions. Nominal loads are equivalent to those provided by the designated UUT and the corresponding ID during normal (no failure) operating conditions. However, the electrical characteristics of interfacing devices may be degraded by such out of tolerance conditions as low power, improper line impedance or signal ground returns, distorted or delayed timing pulses low signal to noise ratios, or extensive operation under extreme environmental conditions. Any new or expanded UUT test application may require performance at or exceeding the interface component manufacturer's specification. For this reason, Table II, which lists typical device specifications, is included to show the design limits for I/O component devices in the IMUTS II interface.

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30. TABLE I - ZIF INTERFACE MODULES

Table I lists the typical interface modules installed in IMUTS II. For convenience, these modules have been given names corresponding with the general test functions.

30.1 Module Characteristics

For each module, the electronic devices which provide the I/O signals are described under the column labeled "FUNCTION." The individual I/O channels are listed under the heading "CH" for each device type. Specific ZIF connector pins shall be assigned to each signal channel, reference, and shield.

30.2 Usage

The "USAGE" column defines the direction of information flow with respect to the IMUTS II. Some devices are bi-directional and have both input and output functions.

30.3 Range

The general impedance, voltage, current, and waveform characteristics are included under the column titled "RANGE." Unless otherwise noted, values listed here are considered typical for all channels of the same device. Unless otherwise noted, impedance is in ohms, voltage is in dc or rms volts, and current is dc or rms amps.

30.4 Remarks

Additional information for each device is provided under the column labeled "REMARKS." Reference is made to Table II which contains component manufacturer's data for each device type.

40. TABLE II - INTERFACE DEVICE CHARACTERISTICS (TYPICAL)

Table II consists of manufacturer's specifications for various I/O devices. The electronic data shown shall be considered typical for the particular device listed. The actual characteristics of each device at the ZIF may vary from the values shown by the manufacturer due to different circuit applications or the substitution of alternate components or manufacturers.

50. TABLE III - DEFINITIONS

Table III defines the logic levels associated with the terms "TTL," "STD TTL," "LOW POWER SCHOTTKY," "CMOS RECEIVER," and "INTERFACE GATE RECEIVER" as used in this specification.

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TABLE I. (INDEX) ZIF INTERFACE MODULES

MODULE NAME	PAGES
1. SDC INTERFACE	39
2. SDC BUFFER	40-46
3. AMUX CH 1	47
4. AMUX CH 2	48
5. MEMORY INTERFACE	49-51
6. DELTA FUNCTION	52-57
7. NO-GO MONITOR	58-65
8 DISCRETE I/O #1	66-70
9. DISCRETE I/O #2	71-75
10 ANALOG SELECT #1	76-77
11. ANALOG SELECT #2	78-79
12. ANALOG SELECT #3	80-81
13. ANALOG SELECT #4	82-83
14. ANALOG/DIGITAL	84
15. SYNC-RES INPUT #1	85-87
16. SYNC-RES INPUT #2	88-90
17. RES/DIGITAL 2 SPD	91-93
18 UUT POWER	92-95
19. POWER CONTROL	96-97
20. A2 ASSEMBLY	98-99
21. CHASSIS	100
22. SIGNAL GRDS	101-102
23 SINS INTERFACE (PROCESSOR)	103
24 DIRECT INSTR ACCESS	104
25. PROGRAMMABLE TERMINATION	105-106
26. CARD TEST CONNECTOR 3 - 4	107-118
27 SPARES	119-120

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS

TEST STATION MODULE / SDC INTERFACE		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TRI-STATE										
TRUE	0		X	A21W	A26A	SERIES 470 IN EACH OUTPUT	TTL			TABLE II (U) DIFFERENTIAL SELF CHECK CLOCK MAY BE USED AS A DISCRETE OR BE SWITCHED AT ANY RATE UP TO 10 KHz PULSE WIDTH IS PROGRAMMABLE TO A MINIMUM OF 50 µs WITH 50 µs RESOLUTION
FALSE	0		X	A21X	A26A		TTL			

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / SDC BUFFER		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BI-DIR. BUS (TYPE A). T^2L REC & OPEN COL. DRIVER WITH 5V PU.	0	X		A25N	A25P		TTL			RECVR/DRIVER TABLE II (I)
	0		X	A25N	A25P					DRIVER TABLE II (B) PU-2 2K to 5V
	1	X	X	A25Q	A25R					DATA SHOULD BE SYNCHRONOUS WITH INTERFACE CLOCK, PAGE 44
	2	X	X	A25S	A25T					REFER TO PAGE 45
	3	X	X	A25U	A25V					
	4	X	X	A25W	A25X					
	5	X	X	A25Y	A25Z					
	6	X	X	A23P	A23N					
	7	X	X	A23R	A23Q					
	8	X	X	A23T	A23S					
	9	X	X	A23V	A23U					
	10	X	X	A23X	A23W					
	11	X	X	A23Z	A23Y					
	12	X	X	B25N	B25P					
	13	X	X	B25Q	B25R					
	14	X	X	B25S	B25T					
	15	X	X	B25U	B25V					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE		USAGE		ZIF CONN J17		RANGE				REMARKS
SDC BUFFER										
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL SCHMITT TRIGGER REC WITH 5V PU										TABLE II (F)
	0	X		B25F	B25E		TTL THRESHOLD HI - 1.4V LO - 5V			2.2K RESISTOR PU TO 5V
	1	X		B25C	B25H					DATA SHOULD BE SYNCHRONOUS WITH INTERFACE CLOCK PAGE 44
	2	X		B25J	B25M					REFER TO PAGE 45
	3	X		B25W	B25X					
	4	X		B25Y	B25Z					
	5	X		B24P	B24N					
	6	X		B24H	B24C					
	7	X		B24K	B24J					
	8	X		B24M	B24L					
	9	X		B24R	B24Q					
	10	X		B24T	B24S					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / SDC BUFFER / FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
TTL DRIVER										TABLE II (D)
	0		X	B24B	B24A		TTL			
	1		X	B24D	B24C					
	2		X	B24F	B24E					
	3		X	A25C	A25D					
	4		X	B24J	B24J					
	5		X	A25F	A25E					
	6		X	A25C	A25H					TABLE II (H) USED ONLY AS A DRIVER
	7		X	A25J	A25K					
	8		X	B25J	B25K					DATA SHOULD BE SYN-CHRONOUS WITH INTERFACE CLOCK PAGE 44
	9		X	B25C	B25D					
	10		X	B25A	B25B					REFER TO PAGE 45

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / SDC BUFFER		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL RECVR										TABLE II (S) SID 116 USED ONLY AS A RECVR RECVR OUTPUTS MAY BE SWITCHED TO DMM OR TIMER/COUNTER VIA SCANNER SCANNER BAND-WIDTH IS APPROX 1 MHz REFER TO PAGE 45
	0	X		A24P	A26A					
	1	X		A24Q	A26A					
	2	X		A24R	A26A					
	3	X		A24G	A26A					
	4	X		A24U	A26A					
	5	X		A24V	A26A					
	6	X		A24W	A26A					
	7	X		A24X	A26A					
	8	X		A24Y	A26A					
	9	X		A24Z	A26A					
	10	X		A24A	A26A					
	11	X		A24B	A26A					
	12	X		A24E	A26A					
	13	X		A24H	A26A					
	14	X		A24F	A26A					
	15	X		A24M	A26A					
	16	X		A24S	A24T					
	17	X		A24C	A24D					
	18	X		A24E	A24G					
	19	X		A24J	A24K					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / SDC BUFFER / FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
DIFFERENTIAL LINE REC WITH 5V PU AND PD RESISTOR ON EACH INPUT										TABLE II (C)
TRUE	0	X		B24V	B24U		DIFF THRESHOLD ± 5V	MAX COMMON MODE VOLTAGE ±5.0V		RESISTORS 560Ω TO 5V AND GRD
FALSE	0	X		B24X	B24W					INTERFACE CLOCK <6 MHz
										REFER TO PAGE 45

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (Cont)

OPERATION OF SDC BUFFER I/O SPECIFICATION 262283

The SDC Buffer is so named because it interfaces with the Signal Data Converter (SDC) portion of the AN/ASN-130 INC. For the purpose of specification 262283, the SDC can be replaced with a UUT which has the necessary characteristics to use the Buffer's interface. The software and hardware link between the Buffer and the test station computer is the SDC interface card which (except for one self-check signal, page 39 of specification) does not connect to the ZIF connector. Subsequent discussion of Buffer ZIF signals implies compatible software and hardware operation of the Buffer with the SDC interface card.

The Buffer interface signals are divided into three functional groups. Programmed I/O, CPU Control (where CPU refers to the CPU in the SDC), and Direct Memory Access (DMA). These signals are interrelated and synchronized with an interface clock which is externally supplied. Refer to page 11 of 262283. The sixteen bi-directional lines (page 40) are used by all three functions for the purpose of transferring address and data information. Functional control lines (page 41) determine whether the bi-directional bus is being used for address or data.

The programmed I/O function will be discussed first. There is one data register associated with the programmed I/O function. It is used to receive data from or transfer data to the CPU. The sequence is initiated under software control by the test station. The SDC buffer driver (page 42, CH 4, B24Z) is then active on the next clock cycle. This generates a CPU interrupt. The detection of an interrupt causes a programmed I/O cycle. This cycle begins with the CPU issuing an I/O address strobe (page 41, CH 6, B24H). This strobe is in synchronism with the clock and causes the bi-directional data to be sampled and decoded. If the programmed I/O decoder detects its address, the programmed I/O proceeds with the cycle.

For a write (into the Buffer) cycle, the data is strobed into the data register during write strobe (page 41, CH 8, B24M) time. This strobe is in synchronism with the clock and has a duration of one clock period. The write strobe occurs a minimum of two clock periods after the address strobe.

For a read (by the CPU) cycle, the data is strobed onto the bi-directional data bus during the read strobe (page 41, CH 7, B24K) time. The read strobe is in synchronism with the clock and has a duration of one clock period. The read strobe occurs a minimum of six clock periods after the address strobe occurs.

At the completion of either a read or write strobe, the test station generates an interrupt and then removes it, and the Buffer/CPU logic is ready for another programmed I/O cycle.

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (Cont)

CPU Control is discussed next. CPU Control has two functions: to start/stop the clock, and to generate a CPU reset signal which produces a synchronous clock halt.

The start/stop cycle uses two signals from the SDC (page 41, Channels 3 and 4, B24W and B25Y) in combination with the test station initiated start/stop signal (page 42, Channel 1, B24D), also known as clock enable. The start signal is initiated by software in the test station, and clock enable goes true one clock period later, thus starting the CPU clock.

The stop signal is initiated by software. Buffer logic monitors the two signals from the SDC. The first signal (page 41, CH 4, B25Y) goes true on the falling edge of the clock. The second signal (page 42, CH 3, B25W) goes true two clock periods later on the falling edge of the clock. The clock enable (page 42, CH 1, B24D) will then go false two clock periods after CH 3 goes true.

The CPU reset signal is initiated by the test station. First the stop sequence as described above occurs, and one millisecond later, a one millisecond pulse is generated (page 42, CH 2, B24F). This is the CPU reset signal.

CPU Control also has two other signals (page 42, Channels 0 and 3, B24B and A25C) that, from the standpoint of the specification, can be considered to be independent (but not simultaneous above 10KHz) sources of TTL square waves up to 20KHz.

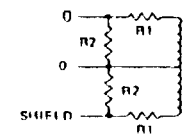
Direct Memory Access (DMA) is discussed next. The DMA function uses the remaining interface signals. The DMA function can be requested by one of three DMA request lines (page 42, Channels 8, 9, or 10) There are three DMA acknowledge signals (page 41, Channels 0, 1, or 5) that are paired with the DMA request signals Test station software selects which DMA request/acknowledge pair is to be used. One pair (page 42, Channel 8, and page 41, Channel 0) will be discussed.

A DMA cycle is initiated by test station software. The DMA request (page 42, Channel 8, B25J) is asserted within two clock periods. The CPU responds with the DMA ACK (Data Memory Access Acknowledge) signal (page 41, Channel 0, B25E) within two clock periods. The DMA ACK signal makes its transition on the positive going edge of the clock. The CPU asserts the DMA CYC (Data Memory Access Cycle) signal (page 41, Channel 2, B25L) a half clock period later. DMA logic asserts and monitors the following signals for one clock period page 41, Channels 9 and 10, B24R and B24T, page 42, Channels 5, 6, and 7; and the address lines (bi-directional bus, page 40, all channels) The DMA cycle will either be a read (from the CPU) or a write (into the CPU) depending on the state of the signals on page 41, Channels 9 and 10 Logic "1" causes read, logic "0" causes write

DMA data (all channels, page 40) transfer occurs a minimum of two clock periods after DMA CYC is asserted. Data transfer lasts for one clock period. The DMA CYC signal goes false at the center of data transfer. The DMA ACK signal goes false one clock period following the data transfer period, and this event terminates the DMA cycle.

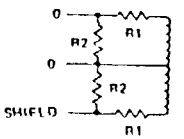
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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / AMUX CH 1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SHORT STUB DIFFERENTIAL DRIVER/RECVR	0	X	X	A20P	A26A	Z IN - SEE REMARKS	20V P-P		BINARY 1 MHZ SERIAL DATA BUSS	INPUT/OUTPUT NETWORK  R1 = 53.6Ω ±1% R2 = 34.8Ω ±1% DATA FORMAT IS PER MCDONNELL DOUGLAS REPORT MDC A3818 REV B THE AMUX HAS AN 1K SERIAL MEMORY WHICH STORES 32 DATA WORDS AND ONE COMMAND WORD MINIMUM ACCESS TIME FOR A 33 WORD TRANSFER (READ/WRITE) IS 1.2 MS
	0*	X	X	A20N	A26A	Z IN - SEE REMARKS	20V P-P		BINARY 1 MHZ SERIAL DATA BUSS	
	1	X	X	A19N	A26A	Z IN - SEE REMARKS	20V P-P		BINARY 1 MHZ SERIAL DATA BUSS	

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / AMUX CH 2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
LONG STUB DIFFERENTIAL DRIVER/RECVR	0	X	X	A18P	A26A	Z IN - SEE REMARKS	20V P-P		BINARY 1 MHZ SERIAL DATA BUSS	INPUT/OUTPUT NETWORK  R1 = 51.6Ω ±1% R2 = 74.8Ω ±1% AMUX CH 2 IS IDENTICAL TO AMUX CH 1 (PAGE 47)
	0*	X	X	A18N	A26A	Z IN - SEE REMARKS	20V P-P		BINARY 1 MHZ SERIAL DATA BUSS	
	S	X	X	A19P	A26A	Z IN - SEE REMARKS	20V P-P		BINARY 1 MHZ SERIAL DATA BUSS	

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE MEMORY I/F		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
DIFF LINE DRIVER										TABLE II (S)
	0		X	B08P	B08N		TTL LEVELS	±40 ma	BINARY	
			X*	B08Q	B08R		TTL LEVELS		BINARY	
	1		X	B08X	B08W		TTL LEVELS	±40 ma	BINARY	
			X*	B08Y	B08Z		TTL LEVELS		BINARY	
	2		X	B08H	B08C		TTL LEVELS	±40 ma	BINARY	
			X*	B08J	B08G		TTL LEVELS		BINARY	
			X	B08C	B08D		TTL	0.4 ma SOURCE AT 2.5V	SQUARE	2.5 MHz TEST POINT CHANNEL 0 PROVIDES A 4 µSEC PULSE AND THEN IS LOW FOR 84 µSEC PLUS TIME REQUIRED FOR 3 BUS TRANSACTIONS. CHANNEL 1 PROVIDES TWENTY-TWO 400 ns PULSES OVER AN 88 µSEC PERIOD. THIS MAY BE REPEATED AFTER 3 BUS TRANSACTIONS. THE FIRST PULSE OCCURS DURING THE CHANNEL 0 SYNC PULSE. CHANNEL 2 PROVIDES A 22 BIT WORD IN 88 µSEC. THIS MAY BE REPEATED AFTER 3 BUS TRANSACTIONS

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE MEMORY I/F		USAGE		ZIF CONN J17		RATE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
DIFF. LINE REC'R										TABLE II (C)
	0	X		B08I	B08K		TTL LEVELS		BINARY	CHANNEL 0 ACCEPTS ONE SYNC PULSE PER 88 µSEC CYCLE AND IGNORES FURTHER PULSE CHANGES
		X*		B08M	B08K		TTL LEVELS		BINARY	
	1	X		B08T	B08S		TTL LEVELS		BINARY	CHANNEL 1 ACCEPTS ONE 22 BIT WORD DURING AN 88 µSEC CYCLE IF NOT READ BY THE PROCESSOR, THIS DATA WOULD BE OVER-WRITTEN BY THE NEXT 22 BIT WORD
		X*		B08U	B08V		TTL LEVELS		BINARY	

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / MEMORY I/F		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
OPEN COLLECTOR DRIVER WITH PU TO +15V										TABLE II (T)
	0		X	B08E	B08F		0 - 15	40 MA MAX		OPEN COLL DRIVER WITH 10K OHM PULLUP TO +15V
	1		X	B08A	B08B		0 - 15	40 MA MAX		OPEN COLL DRIVER WITH 10K OHM PULLUP TO +15V CHANNELS 0 AND 1 CAN CHANGE STATE EVERY 88 µs CHANNELS ARE INDEPENDENT

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DELTA FUNCTION		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
DIFFERENTIAL LINE REC. (SEE REMARKS)										TABLE II (C)
	0	X		B14A	A26A		TTL LEVELS		CONTINUOUS SQ WV	TWISTED PAIRS
	0*	X		B14B	A26A		TTL LEVELS			
	1	X		A14B	A26A		TTL LEVELS		PULSE TRAIN	
	1*	X		A14A	A26A		TTL LEVELS		PULSE TRAIN	
	2	X		A14D	A26A		TTL LEVELS		PULSE TRAIN	
	2*	X		A14C	A26A		TTL LEVELS		PULSE TRAIN	
	3	X		A14F	A26A		TTL LEVELS		PULSE TRAIN	
	3*	X		A14E	A26A		TTL LEVELS		PULSE TRAIN	SEE PAGE 53
	4	X		A14H	A26A		TTL LEVELS		PULSE TRAIN	
	4*	X		A14G	A26A		TTL LEVELS		PULSE TRAIN	
	5	X		A14K	A26A		TTL LEVELS		PULSE TRAIN	
	5*	X		A14J	A26A		TTL LEVELS		PULSE TRAIN	
	6	X		A14M	A26A		TTL LEVELS		PULSE TRAIN	
	6*	X		A14L	A26A		TTL LEVELS		PULSE TRAIN	

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (Cont)

REMARKS
Explanation of Input Characteristics of Channels 0-6*, Page 52
Channels 0 and 0* receive a differential clock. Pulses entering channels 1-6 must be in phase with this clock to be counted properly. There are three independent counters associated with channels 1-6*. Counter number one is tied to channels 1, 1*, 2, and 2*. Counter number two is tied to channels 3, 3*, 4, and 4*, and counter number three is tied to the remaining channels. Counter number one will be discussed, and the discussion is valid for the other counters. A differential pulse train received on channels 1 and 1* will cause the counter to count up A differential pulse train received on channels 2 and 2* will cause the counter to count down. If data is entering channels 1 and 1*, channels 2 and 2* should be inactive (2 LOW, 2* HIGH) If data is entering channels 2 and 2*, channels 1 and 1* should be inactive (1 LOW, 1* HIGH) If these conditions do not pertain, a pulse coincidence error flag will be set on the Delta function card's status control register Channel 1 and 2 count on the negative going transition of a pulse Accurate UP-DOWN counting will occur if the transitions are $\geq$50 ns apart. If the transitions are $\leq$40 ns apart, the pulse coincidence error flag will be set Counter capacity is 2048. There is no overflow sensing. The counter is reset when its contents are "READ" by the processor A "READ" operation requires approximately 800 ns Time between readings is system and application program dependent, and could be in the 60-100 µsec range Card hardware can accurately process data up to a 5 MHz rate

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DELTA FUNCTION		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
DIFFERENTIAL OUTPUT DRIVER										TABLE II (S)
	0		X	A14P	A26A		TTL LEVELS		RECTANGULAR PULSES	TWISTED PAIRS
	0*		X	A14N	A26A		TTL LEVELS		RECTANGULAR PULSES	TWISTED PAIRS
	1		X	A11T	A26A		TTL LEVELS		RECTANGULAR PULSES	TWISTED PAIRS
	1*		X	A14S	A26A		TTL LEVELS		RECTANGULAR PULSES	TWISTED PAIRS
	2		X	A14V	A26A		TTL LEVELS		RECTANGULAR PULSES	TWISTED PAIRS
	2*		X	A14W	A26A		TTL LEVELS		RECTANGULAR PULSES	TWISTED PAIRS
										150 Hz SQ WAVE IS MAX FREQ AT LOW FREQUENCIES, THE WAVESHAPE IS ASSYMETRICAL, WITH ON AND OFF TIMES VARIABLE IN INCREMENTS OF 1/300 SEC UNDER PROGRAM CONTROL UP TO AND INCLUDING FULL ON AND FULL OFF

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DELTA FUNCTION		USACE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL DRIVER										TABLE II (V) TWISTED PAIRS
	0		X	A14Q	A14R		TTL LEVELS	LOW POWER SCHOTTKY		
	1		X	A14U	A14V		TTL LEVELS	LOW POWER SCHOTTKY		
	2		X	A14Y	A147		TTL LEVELS	LOW POWER SCHOTTKY		WAVEFORMS SAME AS CHANNELS 0* 1* AND 2* ON PAGE 54

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DELTA FUNCTION		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL RECVR										TABLE II (W) TWISTED PAIRS
	0	X		B14N	B14P		TTL LEVELS	LOW POWER SCHOTTKY	RECTANGULAR PULSES	
	1	X		B14Q	B14R		TTL LEVELS	LOW POWER SCHOTTKY	RECTANGULAR PULSES	
	2	X		B14S	B14T		TTL LEVELS	LOW POWER SCHOTTKY	RECTANGULAR PULSES	
	3	X		B14U	B14V		TTL LEVELS	LOW POWER SCHOTTKY	RECTANGULAR PULSES	SEE PAGE 57
	4	X		B14W	B14X		TTL LEVELS	LOW POWER SCHOTTKY	RECTANGULAR PULSES	
	5	X		B14Y	B14Z		TTL LEVELS	LOW POWER SCHOTTKY	RECTANGULAR PULSES	

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (Cont)

REMARKS
Explanation of Input Characteristics of Channels 0-5, Page 56
These six channels provide three pairs of inputs to three independent counters. Channels 0 and 1 will be discussed, and the discussion is valid for channels 2 and 3, and for channels 4 and 5. Both channels count on the negative going edge of a pulse. Pulses entering channel 0 will cause the counter to count up. Pulses entering channel 1 will cause the counter to count down. The counter will count accurately up and down if the negative transitions on each channel are $\geq$50 ns apart. If the transitions are $\leq$40 ns apart, the counter will count up, and a pulse coincidence error flag will be set at the Delta function card's status control register. For counting on one channel, the other should be kept in a "HIGH" state. If it is not, the pulse coincidence error flag will be set, but the counting sequence will not be altered. Counter capacity is 255. There is no overflow sensing. The counter is reset when its contents are "READ" by the processor. A "READ" operation requires approximately 800 ns. Time between readings is system and application program dependent, and could be in the 60-100 usec range. Card hardware can accurately process data up to a 5 MHz rate, but the system could not read the counters often enough to prevent overflow at that rate.

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE NO-GO MONITOR		USACF		ZIF CONN J17		PANCF				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BI-DIRECTIONAL TTL DRIVER/CMOS RECEIVER	0	X	X	A04P	A26A	ZIN = 10K	0 TO +15			TABLE II (J) RECVR TABLE II (K) DRIVER BOTH CHANNELS ARE IDENTICAL AND INDEPENDENT. EACH CHANNEL HAS A TABLE II (J) RECEIVER AND A TABLE II (K) DRIVER SIGNALS MAY BE CONSIDERED STATIC. OUTPUTS ARE PROGRAMMABLE AS DISCRETES
	1	X	X	A04T	A26A	ZIN = 10K	0 TO +15			

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / NO-GO MONITOR		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL RECVR WITH 5V PU										TABLE II (L) (M) 1K PU EACH INPUT IS CONNECTED TO THE DEVICES IN TABLE II (L) AND (M)
	0	X		B04T	A26A					
	1	X		B04U	A26A					
	2	X		B04V	A26A					
	3	X		B04W	A26A					
	4	X		B04X	A26A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISITCS (cont)

TEST STATION MODULE / NO-GO MONITOR		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL RECVR										TABLE II (Q) (P) EACH INPUT IS CONNECTED TO THE DEVICES IN TABLE II (Q) AND (R)
	0	X		B04R	A26A					
	1	X		B04S	A26A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE NO-GO MONITOR		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BI-DIR BUS TTL RECVR WITH TRI-STATE DRIVER (CHANNELS 0-13)										RECVR TABLE II (I) (G)
BI-DIR BUS TTL RECVR WITH OPEN COLLECTOR DRIVER (CHANNEL 14)										DRIVER TABLE II (O)
	0	X	X	A04A	A26A					1K PU
	1	X	X	A04B	A26A					1K PU
	2	X	X	A04C	A26A					1K PU
	3	X	X	A04D	A26A					1K PU
	4	X	X	A04E	A26A					1K PU
	5	X	X	A04F	A26A					1K PU
	6	X	X	A04G	A26A					1K PU
	7	X	X	A04H	A26A					1K PU
	8	X	X	A04J	A26A					1K PU
	9	X	X	A04M	A26A					1K PU
	10	X	X	A03N	A26A					1K PU
	11	X	X	A03P	A26A					1K PU
	12	X	X	A03Q	A26A					1K PU
	13	X	X	A03R	A26A					1K PU
	14	X	X	A03T	A26A					RECVR TABLE II (I) (N) DRIVER TABLE II (P) 1K PU EACH INPUT IS CONNECTED TO TWO TYPES OF RECEIVERS OUTPUTS MAY BE PROGRAMMED AS DISCRETES

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / NO-GO MONITOR		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
ANALOG COMPARATORS										TABLE II (X)
	0	X		B04D	B04C&H		THRESHOLD V_T = -8.05V		STATIC	±15 VDC MAX INPUT
	1	X		B04C	B04C&H		THRESHOLD V_T = -8.05V			
	2	X		B04A	B04C&H		THRESHOLD V_T = -8.77V			
	3	X		B04B	B04G&H		THRESHOLD V_T = -8.94V			INPUTS ARE INDEPENDENT
	4	X		B04C&H	B04G&H		GND			RETURN

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / NO-GO MONITOR		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
INTERFACE GATE WITH 15V PU										TABLE II (A)
	0	X		B04Y	A26A		+28 VDC MAX		GND OR +15V	LEVEL SHIFTED INTERFACE GATE 3.3K PU TRIGGER LEVEL +8V
	1	X		B04Z	A26A				GND OR OPEN	

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / NO-GO MONITOR		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL DRIVER	0		X	B04Q	A26A		TTL		1.05 SEC NEGATIVE PULSE	TABLE II (Y) RETRIGGERABLE PULSE IF TRIGGERED MORE OFTEN THAN 1 SEC OUTPUT WILL REMAIN ACTIVE LOW

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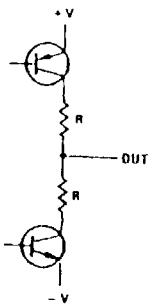
TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / NO-GO MONITOR		USACE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
DISCRETE OUTPUTS										
	0		X	B04N	A26A		+20 ±1V -20 ±1V GND	0 1 AMP MAX		RELAY SWITCHING
	1		X	B04M	A26A		+20 ±1V -20 ±1V GND	0 1 AMP MAX		CHANNELS ARE INDEPENDENT

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE DISCRETE I/O #1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TRI-STATE OUTPUT										SOLID STATE SWITCHING
	0		X	B03W	A26A	1K	±15V DC ±2%	100 mA (MAX)	DC	
	1		X	B03X	A26A	390	±15V DC ±2%			
	2		X	B03Y	A26A	4.7	±10V DC ±3%			



V	R
15	1K
15	390
10	4.7 IN EMITTER NOTHING IN COLLECTOR

MAXIMUM SWITCHING RATE IS 23 KHz SQ WAVE

CHANNELS ARE INDEPENDENT

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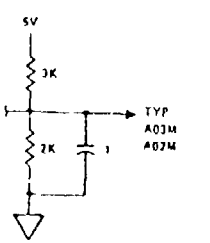
TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE DISCRETE I/O #1		USACE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RELAY OUTPUT										CONTACT RATING 1 AMP RESISTIVE AT 28 VDC 0.2 AMP INDUCTIVE 0.32H at 28 VDC OPERATE TIME = 10 ms RELAYS MAY BE INDEPENDENTLY OPERATED
COM	0		X	B03A	NONE					
NO	0		X	B03C	NONE					
NC	0		X	B03B	NONE					
COM	1		X	B03D	NONE					
NO	1		X	B03F	NONE					
NC	1		X	B03E	NONE					
COM	2		X	B03G	NONE					
NO	2		X	B03J	NONE					
NC	2		X	B03H	NONE					
COM	3		X	B03K	NONE					
NO	3		X	B03M	NONE					
NC	3		X	B03L	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE DISCRETE I/O #1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BI-DIRECT I/O										AS A
OPEN COLLECTOR DRIVER	1		X	A03A	A26A		+30V DC (MAX)	SINK 32 mA (MAX)	DC	TABLE II (E) DRIVER
HI INPUT IMP RECEIVER	1	X			A26A	1 MEG (MIN)				TABLE II (R) RCVR
	2	X	X	A03B	A26A					INPUT = GREATER THAN REF
	3	X	X	A03C	A26A					FALSE = LESS THAN REF
GROUP #1	4	X	X	A03D	A26A					I/O RATE LIMITED TO 23 KHz SEE PWR CONTROL PAGE 59
	5	X	X	A03E	A26A					
	6	X	X	A03F	A26A					CHANNELS 1-16 ARE PROGRAMMED AS A GROUP AS DRIVERS OR RECEIVERS
	7	X	X	A03G	A26A					
	8	X	X	A03H	A26A					
	9	X	X	A02A	A26A					
	10	X	X	A02B	A26A					
	11	X	X	A02C	A26A					
	12	X	X	A02D	A26A					
GROUP #2	13	X	X	A02E	A26A					
	14	X	X	A02F	A26A					
	15	X	X	A02G	A26A					
	16	X	X	A02H	A26A					
GROUP 1 REF		X		A03M	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE *
GROUP 2 REF		X		A02M	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE *

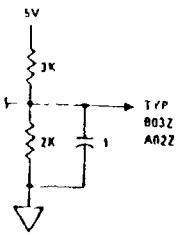


*FOR RECEIVER/COMPARATOR

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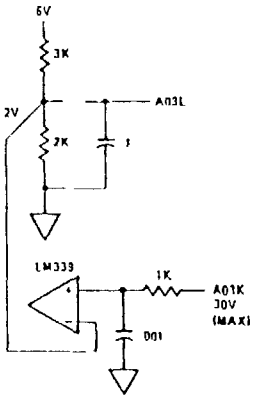
TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE DISCRETE I/O #1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BI-DIRECT I/O OPEN COLLECTOR DRIVER	17		X	B03N	A26A		+30V DC (MAX)	SINK 32 mA (MAX)	DC	TABLE II (F)
HI INPUT IMP RECEIVER	17	X			A26A	1 MEG (MIN)				TABLE II (R)
	18	X	X	B03P	A26A					TRUE = GREATER THAN REF
GROUP #3	19	X	X	B03Q	A26A					FALSE = LESS THAN REF
	20	X	X	B03R	A26A					CHANNELS 17-32 ARE PROGRAMMED AS A GROUP AS DRIVERS OR RECEIVER INDEPENDENTLY OF CHANNELS 1-16
	21	X	X	B03S	A26A					
	22	X	X	B03T	A26A					
	23	X	X	B03U	A26A					
	24	X	X	B03V	A26A					I/O RATE CHANNELS 17-32 IS SAME AS CHANNELS 1-16
	25	X	X	A02N	A26A					
	26	X	X	A02P	A26A					
	27	X	X	A02Q	A26A					
GROUP #4	28	X	X	A02R	A26A					
	29	X	X	A02S	A26A					
	30	X	X	A02T	A26A					
	31	X	X	A02U	A26A					
	32	X	X	A02V	A26A					
GROUP #3 REF		X		B03Z	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE
GROUP #4 REF		X		A02Z			+2V DC (NOM)			SELECTABLE DC REFERENCE



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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DISCRETE I/O #1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BROADSIDE STROBE FOR ALL 32 BI-DIRECT I/O										
STROBE		X		A03K	A03J	1 MEG (MIN)	+30V DC (MAX)			TABLE II (R) TRUE = GREATER THAN REF FALSE = LESS THAN REF
STROBE REF.		X		A03L	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE  STROBE SHOULD BE ≥ 4μsec WIDTH 100 μsec MINIMUM BETWEEN STROBES

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE DISCRETE I/O #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TRI-STATE OUTPUT										
	0		X	B02J	A26A	1K	±15V DC ±2%	100 mA (MAX)	DC	SOLID STATE SWITCHING THIS MODULE IS IDENTICAL TO DISCRETE I/O #1
	1		X	B02K	A26A	390	±15V DC ±2%			
	2		X	B02L	A26A	10 (MAX)	±10V DC ±3%			

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DISCRETE I/O #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RELAY OUTPUT										CONTACT RATING
COM	0		X	A01N	NONE		500 VDC (MAX)			1 AMP RESISTIVE AT 28 VDC
NO	0		X	A01O	NONE					0.2 AMP INDUCTIVE 0.32H AT 28 VDC
NC	0		X	A01P	NONE					
COM	1		X	A01R	NONE					
NO	1		X	A01T	NONE					
NC	1		X	A01S	NONE					
COM	2		X	A01U	NONE					
NO	2		X	A01W	NONE					
NC	2		X	A01V	NONE					
COM	3		X	A01X	NONE					
NO	3		X	A01Z	NONE					
NC	3		X	A01Y	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE DISCRETE I/O #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BI-DIRECT I/O OPEN COLLECTOR DRIVER	1		X	B02N	A26A		+30V DC (MAX)	SINK 32 mA (MAX)	DC	TABLE II (F)
HI INPUT IMP RECEIVER GROUP #1	1	X				1 MEG (MIN)				TABLE II (P)
	2	X	X	B02P	A26A					TRUE = GREATER THAN REF FALSE = LESS THAN REF
	3	X	X	B02Q	A26A					
	4	X	X	B02R	A26A					
	5	X	X	B02S	A26A					
	6	X	X	B02T	A26A					
	7	X	X	B02U	A26A					
	8	X	X	B02V	A26A					
GROUP #2	9	X	X	B01N	A26A					
	10	X	X	B01P	A26A					
	11	X	X	B01Q	A26A					
	12	X	X	B01R	A26A					
	13	X	X	B01S	A26A					
	14	X	X	B01T	A26A					
	15	X	X	B01U	A26A					
	16	X	X	B01V	A26A					
GROUP #1 REF		X		B02Z	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE
GROUP #2 REF		X		B01Z	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE DISCRETE I/O #2		USAGE		ZIF CONN J17		PALCE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BI-DIRECT I/O OPEN COLLECTOR DRIVER										TABLE II (E)
HI INPUT IMP RECEIVER	17		X	B02A	A26A		+30V DC (MAX)	SINK 32 mA (MAX)	DC	
	17	X			A26A	1 MEG (MIN)				TABLE II (B)
GROUP #3	18	X	X	B02B	A26A					TRUE = GREATER THAN REF FALSE = LESS THAN REF
	19	X	X	B02C	A26A					
	20	X	X	B02D	A26A					
	21	X	X	B02E	A26A					
	22	X	X	B02F	A26A					
	23	X	X	B02G	A26A					
	24	X	X	B02H	A26A					
	25	X	X	A01A	A26A					
GROUP #4	26	X	X	A01B	A26A					
	27	X	X	A01C	A26A					
	28	X	X	A01D	A26A					
	29	X	X	A01E	A26A					
	30	X	X	A01F	A26A					
	31	X	X	A01G	A26A					
	32	X	X	A01H	A26A					
GROUP #3 REF.		X		B02H	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE
GROUP #4 REF		X		A01H	A26A		+2V DC (NOM)			SELECTABLE DC REFERENCE

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DISCRETE I/O #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
BROADSIDE STROBE FOR ALL 32 BI-DIRECT I/O										
STROBE		X		B02X	B02W	1 MEG (MIN)	+30V DC (MAX)			TABLE II (P) TRUE = GREATER THAN REF FALSE = LESS THAN REF
STROBE REF		X		B02Y	A26A	+2V DC (NOM)				SELECTABLE DC REFERENCE

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG SELECT #1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 0										
	0	X		B21O	NONE	ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX	DC DISCRETES, SINE, RAMP, PULSE 100 ns MIN APPLIES TO RESPONSE GROUPS 0-7	THIS GROUP OF INPUTS ARE THE INPUTS TO A 32 1 MUX THE OUTPUT GOES TO INTERNAL BUSS NO 0 THIS BUSS CAN BE SWITCHED TO THE FOLLOWING DMM RATIO H, L, +I -I COUNTER/TIMER EXT GATE, CHA CHB R/D CONV COS LO COUNTER/TIMER FREQ 0-1 MHz FREQ RATIO 0-1 MHz PERIOD 0-1 MHz PERIOD AVG 1 TO 10^8 PERIODS TIME INTERVAL 0.1 µS TO 10^9 s TOTALIZE 0-1 MHz CHANNELS A AND B 200 mv TO 250 V RMS 0-1 MHz MIN PULSE WIDTH 100 ns (REFER TO PAGE 40 FOR DAA CHARACTERISTICS VIA ANALOG SCANNER)
	1	X		A22C						
	2	X		B21R						
	3	X		A22D						
	4	X		B21S						
	5	X		A22F						
	6	X		B21T						
	7	X		A22F						
	8	X		B21U						
	9	X		A22C						
	10	X		B21V						
	11	X		A22H						
	12	X		A22K						
	13	X		B21Y						
	14	X		A22L						
	15	X		B21Z						
	16	X		A21Q						
	17	X		B22C						
	18	X		A21R						
	19	X		B22D						
	20	X		A21S						
	21	X		B22E						
	22	X		A21T						
	23	X		B22F						
	24	X		A21U						
	25	X		B22G						
	26	X		A21V						
	27	X		B22H						
	28	X		B22K						
	29	X		A21Y		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	30	X		B22L						
	31	X		A21Z	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG SELECT #1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 1										
	0	X		B21C	NONE	ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		THIS GROUP OF INPUTS ARE THE INPUTS TO A 32 1 MUX THE OUTPUT GOES TO INTERNAL BUSS 10 1 THIS BUSS CAN BE SWITCHED TO THE FOLLOWING D 1 RATIO H F +I -I COUNTER TIMER EXT GATE CHA CHB R/D CONV COS HI S/D CONV Z DMM 30 HZ TO 100 KHz 0 TO ±500 VDC AC ACCURACY ±2% OF INPUT DC ACCURACY ±0.02% OF INPUT OHM (100 OHM RANGE TO 10 MEGOHM RANGE) ±0.016% OF INPUT 4 WIRE
	1	X		A22Q						
	2	X		B21D						
	3	X		A22R						
	4	X		B21E						
	5	X		A22S						
	6	X		B21F						
	7	X		A22T						
	8	X		B21G						
	9	X		A22U						
	10	X		B21H						
	11	X		A22V						
	12	X		A22X						
	13	X		B21J						
	14	X		A22Y						
	15	X		B21M						
	16	X		A21C						
	17	X		B22Q						
	18	X		A21D						
	19	X		B22P						
	20	X		A21F						
	21	X		B22S						
	22	X		A21E						
	23	X		B22T						
	24	X		A21C						
	25	X		B22U						
	26	X		A21H						
	27	X		B22V						
	28	X		B22X						
	29	X		A21I		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	30	X		B22Y						
	31	X		A21M	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG SELECT #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 2										
	0	X		B19Q	NONE	ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		THIS GROUP OF INPUTS ARE THE INPUTS TO A 32 1 MUX THE OUTPUT GOES TO INTERNAL BUSS NO 2 THIS BUSS CAN BE SWITCHED TO THE FOLLOWING DMM RATIO H L +I -I COUNTER/TIMER EXT. GATE, CHA CHB R/D CONV SIN HI S/D CONV Y
	1	X		A20C						
	2	X		B19R						
	3	X		A20D						
	4	X		B19S						
	5	X		A20F						
	6	X		B19T						
	7	X		A20F						
	8	X		B19U						
	9	X		A20G						
	10	X		B19V						
	11	X		A20H						
	12	X		A20K						
	13	X		B19Y						REFER TO PAGES 76 & 77 FOR ADDITIONAL DATA
	14	X		A20I						
	15	X		B19Z						
	16	X		A19Q						
	17	X		B20C						
	18	X		A19R						
	19	X		B20D						
	20	X		A19S						
	21	X		B20F						
	22	X		A19T						
	23	X		B20F						
	24	X		A19U						
	25	X		B20G						
	26	X		A19V						
	27	X		B20H						
	28	X		B20K						
	29	X		A19Z		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	30	X		B20I						
	31	X		A19Z	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ANALOG SELECT #2		USACE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 3										
	0	X		B19C		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		THIS CIRCUIT OF INPUTS ARE THE INPUTS TO A 32 1 MUX THE OUTPUT GOES TO INTERNAL BUSS NO 3 THIS BUSS CAN BE SWITCHED TO THE FOLLOWING DMM RATIO H L 4I -I COUNTER TIMER EXT GATE CHA CHB R/D CONV SIN LO S/D CONV X REFER TO PAGES 76 & 77 FOR ADDITIONAL DATA
	1	X		A20Q						
	2	X		B19D						
	3	X		A20R						
	4	X		B19F						
	5	X		A20S						
	6	X		B19F						
	7	X		A20T						
	8	X		B19G						
	9	X		A20U						
	10	X		B19H						
	11	X		A20V						
	12	X		A20X						
	13	X		B19L						
	14	X		A20Y						
	15	X		B19M						
	16	X		A19C						
	17	X		B20O						
	18	X		A19D						
	19	X		B20P						
	20	X		A19F						
	21	X		B20S						
	22	X		A19F						
	23	X		B20T						
	24	X		A19G						
	25	X		B20U						
	26	X		A19H						
	27	X		B20V						
	28	X		B20X						
	29	X		A19I		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	30	X		B20Y						
	31	X		A19M						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG SELECT #3		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 4	0	X		B17Q		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		THIS GROUP OF INPUTS ARE THE INPUTS TO A 32 1 MUX THE OUTPUT GOES TO INTERNAL BUSS NO 4 THIS BUSS CAN BE SWITCHED TO THE FOLLOWING DAM RATIO II I, +I, -I COUNTER/TIMER EXT GATE CHA, CHB R/D CONV COS IO REFER TO PAGES 76 & 77 FOR ADDITIONAL DATA
	1	X		A18C						
	2	X		B17R						
	3	X		A18D						
	4	X		B17S						
	5	X		A18F						
	6	X		B17T						
	7	X		A18F						
	8	X		B17U						
	9	X		A18C						
	10	X		B17V						
	11	X		A18H						
	12	X		A18K						
	13	X		B17Y						
	14	X		A18I						
	15	X		B17Z						
	16	X		A17Q						
	17	X		B18C						
	18	X		A17P						
	19	X		B18D						
	20	X		A17S						
	21	X		B18F						
	22	X		A17T						
	23	X		B18F						
	24	X		A17U						
	25	X		B18C						
	26	X		A17V						
	27	X		B18H						
	28	X		B18K						
	29	X		A17Y		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	30	X		B18I						
	31	X		A17Z						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG SELECT #3		USAGE		ZIF CONN J17		PANEL				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 5	0	X		B17C	NONE	ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		THIS GROUP OF INPUTS ARE THE INPUTS TO A 32 1 MUX. THE OUTPUT GOES TO INTERNAL BUS 0 5 THIS BUSS CAN BE SWITCHED TO THE FOLLOWING DMM RATIO H L +I, -I COUNTER/TIMER EXT GATE CHA, CHB R/D CONV COS HI S/D CONV 7 REFER TO PAGES 76 & 77 FOR ADDITIONAL DATA
	1	X		A18Q						
	2	X		B17D						
	3	X		A18R						
	4	X		B17E						
	5	X		A18S						
	6	X		B17F						
	7	X		A18T						
	8	X		B17G						
	9	X		A18U						
	10	X		B17H						
	11	X		A18V						
	12	X		A18X						
	13	X		B17J						
	14	X		A18Y						
	15	X		B17K						
	16	X		A17C						
	17	X		B18Q						
	18	X		A17D						
	19	X		B18R						
	20	X		A17F						
	21	X		B18S						
	22	X		A17F						
	23	X		B18T						
	24	X		A17G						
	25	X		B18U						
	26	X		A17H						
	27	X		B18V						
	28	X		B18X						
	29	X		A17J						
	30	X		B18Y		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	31	X		A17M	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG SELECT #4		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 6										
	0	X		B15Q	NONE	ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		THIS GROUP OF INPUTS ARE THE INPUTS TO A 32 1 MUX THE OUTPUT GOES TO INTERNAL BUSS NO 6 THIS BUSS CAN BE SWITCHED TO THE FOLLOWING DMM RATIO, H L +I, -I COUNTER/TIMER EXT GATE CHA CHB R/D CONV SIN HI S/D CONV Y
	1	X		A16C						
	2	X		B15R						
	3	X		A16D						
	4	X		B15S						
	5	X		A16E						
	6	X		B15T						
	7	X		A16F						
	8	X		B15U						
	9	X		A16G						
	10	X		B15V						
	11	X		A16H						REFER TO PAGES 76 & 77 FOR ADDITIONAL DATA
	12	X		A16K						
	13	X		B15Y						
	14	X		A16I						
	15	X		B15Z						
	16	X		A15Q						
	17	X		B16C						
	18	X		A15R						
	19	X		B16D						
	20	X		A15S						
	21	X		B16E						
	22	X		A15T						
	23	X		B16F						
	24	X		A15U						
	25	X		B16G						
	26	X		A15V						
	27	X		B16H						
	28	X		B16K		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	29	X		A15Z						
	30	X		B16I						
	31	X		A15Z	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG SELECT #4		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESPONSE GROUP 7										
	0	X		B15C	NONE	ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		THIS GROUP OF INPUTS ARE THE INPUTS TO A 32 1 MUX THE OUTPUT GOES TO INTERNAL BUSS NO 7 THE BUSS CAN BE SWITCHED TO THE FOLLOWING DM4 RATIO H L +I -I COUNTER/TIMER EXT GATE CHA CHB P/D CONV SIN LO S/D CONV X REFER TO PAGES 76 & 77 FOR ADDITIONAL DATA
	1	X		A16Q						
	2	X		B15D						
	3	X		A16R						
	4	X		B15F						
	5	X		A16S						
	6	X		B15F						
	7	X		A16T						
	8	X		B15G						
	9	X		A16U						
	10	X		B15H						
	11	X		A16V						
	12	X		A16X						
	13	X		B15I						
	14	X		A16V						
	15	X		B15M						
	16	X		A15C						
	17	X		B16Q						
	18	X		A15D						
	19	X		B16R						
	20	X		A15F						
	21	X		B16S						
	22	X		A15F						
	23	X		B16T						
	24	X		A15G						
	25	X		B16U						
	26	X		A15H						
	27	X		B16V						
	28	X		B16X						
	29	X		A15I		ZIN IS DEPENDENT ON TEST INSTRUMENTS CONNECTED	500V DC MAX	2A MAX 100 VA MAX		
	30	X		B16Y						
	31	X		A15M	NONE					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / ANALOG/DIGITAL		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
A/D CONVERTER MUX INPUT										
HI	4	X		B13A	N/A	10 MEG (MIN)	±10V DC MAX	DC	DC	CONVERTER RESOLUTION IS 11 BITS RELATIVE ACCURACY IS ±0 025% OF FULL SCALE DEVICE CONVERSION TIME IS 28 µSEC SYSTEM APPLICATION LIMITS SUCCESSIVE READINGS OF SAME CHANNEL TO ≥50 µSEC
LO	4	X		B12A						
HI	5	X		B13B						
LO	5	X		B12B						
HI	6	X		B13C						
LO	6	X		B12C						
HI	7	X		B13D						
LO	7	X		B12D						
HI	8	X		B13M						
LO	8	X		B12M						
HI	9	X		B13L						
LO	9	X		B12L						
HI	10	X		B13K						
LO	10	X		B12K						
HI	11	X		B13J						
LO	11	X		B12J						
HI	12	X		B13H						
LO	12	X		B12H						
HI	13	X		B13G						
LO	13	X		B12G						
HI	14	X		B13F						
LO	14	X		B12F						
HI	15	X		B13E						
LO	15	X		B12E	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE		USAGE		ZIF CONN J17		RANGE				REMARKS
SYNC-RES INPUT #1										
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SIGNAL INPUT										COS LO
	0	X		B23F	N/A	500K (MIN) BALANCED	11 8V RMS*		400 HZ	Z COS HI
	1	X		B23G	N/A					Y SIN HI
	2	X		B23F	N/A					X SIN LO
	3	X		B23H	N/A					*INPUT VOLTAGE LEVEL IS PROGRAMMABLE (SELECT 1 OF 3)
										3 4V (3 1-3 7)
										6 2V (5 6-6 8)
										13 0V (10 6-1, 3)
										ACCURACY - 0 020°
										RESOLUTION = 0 0055°
										TRACKING RATE FOR FULL ACCURACY IS 360 DEG/SEC MAX

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TABLE I. ZIF CONNECTOR SIGNAL, INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE		USAGE		ZIF CONN J17		RANGE				REMARKS
SYNC-RES INPUT #1										
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
REFERENCE VOLTAGE										
HI	0	X		B23J	N/A	40K (MIN)	10 TO 135V RMS 26V RMS (NOM)		400 HZ	SYNCHRO REF
LO	0	X		B23K	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / SYNC-RES INPUT #1		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
EXTERNAL LOAD										COS LO Z COS HI Y SIN HI X SIN LO THESE POINTS ARE ALSO COMMON TO SYNC-RES #2 EXTERNAL LOAD OUTPUTS AND ARE USED FOR SELF CHECK ONLY
	0	X		B23A	N/A					
	1	X		B23C	N/A					
	2	X		B23B	N/A					
	3	X		B23D	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE SYNC-RES INPUT #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SIGNAL INPUT										
	0	X		B01H	N/A	500K (MIN) BALANCED	11 8V RMS*		400 HZ	COS LO Z COS HI Y SIN HI X SIN LO *INPUT VOLTAGE LEVEL IS PROGRAMMABLE (SELECT 1 OF 3) 3 4V (3 1-3 7) 6 2V (5 6-6 8) 13 0V (10 6-14 3) ACCURACY - 0 020° RESOLUTION = 0 0055° TRACKING RATE FOR FULL ACCURACY IS 360 DEG/SEC MAX
	1	X		B01F						
	2	X		B01G						
	3	X		B01E						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE SYNC-RES INPUT #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
REFERENCE VOLTAGE										
HI	0	X		B01D	N/A	40K (MIN)	10 TO 135V RMS 26V RMS (NOM)		400 HZ	SYNCHRO REF
LO	0	X		B01C	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / SYNC-RES INPUT #2		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
EXTERNAL LOAD										
	0	X		B23A	N/A					COS LO
	1	X		B23C	N/A					Z, COS HI
	2	X		B23B	N/A					Y SIN HI
	3	X		B23D	N/A					X, SIN LO
										THESE POINTS ARE ALSO COMMON TO SYNC-RES #1 EXTERNAL LOAD OUTPUTS AND ARE USED FOR SELF CHECK ONLY

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE RES/DIGITAL 2 SPD		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESOLVER INPUT										COARSE (RATIO 1 x 8)
CHANNEL 1	0	X		B07R	N/A	46K (MIN) BALANCED	11 8V RMS		360 TO 1000 HZ SINUSOID	SIN HI
	1	X		B07S						SIN LO
	2	X		B07T						COS HI
	3	X		B07U						COS LO
										ACCURACY = 0 0041° RESOLUTION = 0 0014°
										FINE (RATIO 1 x 32)
	4	X		B07V						SIN HI
	5	X		B07W						SIN LO
	6	X		B07X						COS HI
	7	X		B07Y	N/A					COS LO
										ACCURACY = 0 0014 RESOLUTION = 0 00017° TRACKING RATE FOR FULL ACCURACY IS 225 DEG/SEC MAX

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE RES/DIGITAL 2 SPD		USAGE		ZIF CONN J17		PANEL				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
RESOLVER INPUT CHANNEL 2										COARSE (RATIO 1 x 8)
	0	X		B07D	N/A	46K (MIN) BALANCED	11.8V RMS		360 TO 1000 HZ SINUSOID	SIN HI
	1	X		B07F						SIN LO
	2	X		B07E						COS HI
	3	X		B07C						COS LO
										FINE (RATIO 1 x 32)
	4	X		B07H						SIN HI
	5	X		B07J						SIN LO
	6	X		B07K						COS HI
	7	X		B07L	N/A					COS LO
										ACCURACY - SAME AS CHANNEL 1

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE RES/DIGITAL 2 SPD		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
REFERENCE VOLTAGE										
AC HI	0	X		B07N	N/A	240K DIFFERENTIAL	4 TO 130V RMS		360 TO 1000 HZ	RESOLVER REF
AC LO	0	X		B07P	N/A					
VEL	1	X		B07Q	A26A	SELECT	8 VDC (NOM)			RESISTOR SELECTED IN I D TO SET THE ANGULAR VELOCITY ERROR LEVEL (DC)

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / UUT POWER FUNCTION		CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
AC POWER	ØA	0		X	B15B	N/A		118 ±7V RMS*	10 AMP/PH (MAX)	400 ±20 HZ SINUSOIDAL	*LINE TO NEUTRAL PHASE A
		0		X	B17B						
		0		X	B19B						
		0		X	B21B						
	ØB	1		X	B15A						PHASE B
		1		X	B17A						
		1		X	B19A						
		1		X	B21A						
	ØC	2		X	B16B						PHASE C
		2		X	B18B						
		2		X	B20B						
		2		X	B22B						
	ØN	3		X	B16A						NEUTRAL
		3		X	B18A						
		3		X	B20A						
		3		X	B22A	N/A					
	26V HI	4		X	A16B	A16A		26V RMS ±5%	1 1 AMP (MAX)		SYNCHRO-RESOLVER REFERENCE
	26V LO	4		X	A16A	N/A					REF RETURN
	CHASSIS	5		X	B17N	N/A		-	-	-	CHASSIS GROUND

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TABLE 1. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / UUT POWER		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
DC POWER										
+28V HI	0		X	A17B	N/A		+28V DC*	15 AMPS (MAX)	DC	*DC OUTPUT VOLTAGE IS PROGRAMMABLE 0 TO 40V DC RESOLUTION 9 8 mv
	0		X	A18B						
	0		X	A19B						
	0		X	A20B						
	0		X	A21B						
+28V LO	1		X	A17A						+28V DC RETURN
	1		X	A18A						
	1		X	A19A						
	1		X	A20A						
	1		X	A21A						
-28V HI	2		X	B20P			-28V DC**	5 AMPS (MAX)	DC	**PROGRAMMABLE 0 TO -40V DC RESOLUTION 9 8 mv
	2		X	B21P						
-28V LO	3		X	B20N						-28V DC RETURN
	3		X	B21N						
REMOTE SENSE +28V HI	4	X		A22A						
REMOTE SENSE +28V LO	5	X		A22B						
REMOTE SENSE -28V HI	6	X		B22P						
REMOTE SENSE -28V LO	7	X		B22N	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / POWER CONTROL		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
OPEN COLLECTOR DRIVER WITH 7.5K PU TO +28V										
	0		X	B01A	A26A		0 TO 28V DC			TABLE II (1)
	1		X	B01B	A26A					MAXIMUM SWITCHING RATE IS APPROXIMATELY 23 KHz UNDER DEDICATED PROGRAM CONTROL. PULSE WIDTH FOR 50% DUTY CYCLE IS 21.7 μSEC

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / POWER CONTROL		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TTL RECVR WITH 1K PU TO 5V	0	X		B01J	A26A		TTL			TABLE II (N) ID INTERLOCK

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION ASSY / A2 ASSY		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
POWER CONTROL FOR PULL UP POWER	0	X		B01F	A26A		0 TO 28V	2 MA AT 0 VDC		A GROUND APPLIES PU POWER

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION ASSY / A2 ASSY		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
PULL UP POWER										
+5V DC	0		X	A22P	A26A		+5V DC 5%	1 AMP MAX	DC	
+15V DC	1		X	A21N	A26A		+15V DC 5%	1 AMP MAX		
+28V DC	2		X	A21P	A26A		+28V DC 5%	1 AMP MAX		

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / CHASSIS		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
POWER LOADS										
	1			B19P	B19N	115Ω 1% 120W				
	2			B18P	B18N	13.3Ω 1% 120W				

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE SEE REMARKS		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSMB SIGNAL GROUNDS				A26A A26B A26C A26D A26P A26F A26G A26H A26J A26K A26L A26M A26N A26P A26Q A26R A26S A26T A26U A26V A26W A26X A26Y A26Z	N/A ↓ N/A					THESE GROUNDS RETURN DIRECTLY TO THE CARD CAGE GROUND PLANE VIA FOUR 1 2 INCH WIDE CONDUCTORS

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE SEE REMARKS		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSMB SIGNAL GROUNDS (cont)				B26A B26B B26C B26D B26E B26F B26G B26H B26J B26K B26L B26M B26N B26P B26Q B26R B26S B26T B26U B26V B26W B26X B26Y B26Z	N/A N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

PROCESSOR MODULE / SINS INTERFACE		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SINS DATA LINK INTERRUPT										TABLE II (c)
TRUE	0		X	B16N	A26A		TTL			DIFFERENTIAL
FALSE	0		X	B16P	A26A		TTL			SINS DATA LINK INTERRUPTS ARE GENERATED EVERY 16 MS AT EACH INTERRUPT A POSITIVE 150 μSEC PULSE IS GENERATED ON B16N AND ITS COMPLEMENT IS GENERATED ON B16P WHEN SINS IS ABSENT THESE LINES ARE INACTIVE (B16N LOW B16P HIGH)

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / DIRECT INSTR ACCESS		USAGE		ZIF CONN. J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
DMM										
-RATIO		X		A21J	N/A		500V DC MAX		DC, DISCRETES, SINE (30 HZ TO 100 KHz)	DMM 30 Hz-100 KHz 0 ±500 VDC
+IN		X		A17P						AC ACCURACY ±1% OF INPUT
-IN		X		A16N						DC ACCURACY ±0 01% OF INPUT
ANALOG GRD		X		A21K			500V DC MAX			OHMS (100 OHM RANGE TO 10 MEGOHM RANGE) ±0 016% OF INPUT 4 WIRE
COUNTER									SINE, RAMP, PULSE, DISCRETES	COUNTER/TIMER FREQ 0-2 MHz
CH A		X		A15N			250V RMS MAX			FREQ RATIO 0-2 MHz
CH B		X		A15P						PERIOD 0-2 MHz
TRIG		X		A16P	N/A		250V RMS MAX			PERIOD AVG 1 TO 10^8 PERIOD
										TIME INTERVAL 0 1 µS TO 10^9S
										TOTALIZE 0-2 MHz
										CHANNELS A AND B 150 MV TO 250 V RMS 0-2 MHz
										MIN PULSE WIDTH 80 ns

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / PROGRAMMABLE TERMINATION FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
DMM LOAD ACCESS										THESE CHANNELS ARE DIRECT ACCESS TO THE LOW SIDE OF THE DMM LOAD WHICH IS LOCATED ON THE PROGRAMMABLE TERMINATION CARD
	0	X		B21X	N/A	0 TO 120K	500V DC MAX		ANY	
	1	X		B21W	N/A		500V DC MAX		ANY	
	2	X		B22W	N/A		500V DC MAX		ANY	

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / PROGRAMMABLE TERMINATION		USACF		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
4.8 KHZ NOTCH FILTER		X		B21J	B21K	>1K	15V RMS		SINE	USE TO DETERMINE HARMONIC CONTENT IN A 4 8 KHZ SINE WAVE

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4 FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
TSM TEST CONNECTOR 3-4 SECTION A										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A1				A08H	N/A					
B1				A08I						
C1				A08J						
D1				B09H						
E1				B09L						
F1				B09K						
H1				B09J						
J1				B09H						
K1				B09C						
L1				B09F						
M1				B09E						
N1				B09D						
P1				B09C						
R1				B09B						
S1				A09A						
T1				NONP						
U1				A09C						
V1				A08F	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4 FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
TSM TEST CONNECTOR 3-4 SECTION A										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A2				NONP	N/A					
B2				A08K						
C2				NONE						
D2				B09Z						
E2				B09Y						
F2				B09X						
H2				B09W						
J2				B09V						
K2				B09U						
L2				B09T						
M2				B09S						
N2				B09R						
P2				B09Q						
R2				B09P						
S2				B09N						
T2				A08H						
U2				A08F						
V2				A08D	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USACF		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION B										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A1				A08Z	N/A					
B1				A08Y						
C1				A08X						
D1				A08W						
E1				A08V						
F1				A08U						
H1				A08T						
J1				A08S						
K1				A08R						
L1				A08Q						
M1				A08P						
N1				NONE						
P1										
R1										
S1										
T1										
U1										
V1				NONE	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION B										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A2				NONE	N/A					
B2				A09M						
C2				A09L						
D2				A09K						
E2				A09J						
F2				A09H						
H2				A09G						
J2				A09F						
K2				A09E						
L2				A09D						
M2				A09C						
N2				NONE						
P2										
R2										
S2										
T2										
U2										
V2				NONE	N/A					

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TABLE I. ZIF CONNECTOP SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION C										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A1				A07N	N/A					
B1				A07P						
C1				A07Q						
D1				A06A						
E1				A06B						
F1				A06C						
H1				A06D						
J1				A06E						
K1				A06F						
L1				A06G						
M1				A06H						
N1				A06J						
P1				A06K						
R1				A06L						
S1				A06M						
T1				NONE						
U1				A06Q						
V1				A06R	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4 FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
TSM TEST CONNECTOR 3-4 SECTION C										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A2				NONE	N/A					
B2				A06N						
C2				NONE						
D2				A04N						
E2				A04P						
F2				A04Q						
H2				A04R						
J2				A04S						
K2				A04T						
L2				A04U						
M2				A04V						
N2				A04W						
P2				A04X						
R2				A04Y						
S2				A04Z						
T2				A06P						
U2				A07R						
V2				A07S	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION D										CARD CAGE POSITION 3 4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A1				A06S	N/A					
B1				A07T						
C1				A07U						
D1				A06N						
E1				A06P						
F1				A06Q						
H1				A06R						
J1				A06S						
K1				A06T						
L1				A06U						
M1				A06V						
N1				A06W						
P1				A06X						
R1				A06Y						
S1				A06Z						
T1				NONE						
U1				A07V						
V1				A07W	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION D										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A2				NONE	N/A					
B2				A06T						
C2				NONE						
D2				B05A						
E2				B05B						
F2				B05C						
H2				B05D						
J2				B05E						
K2				B05F						
L2				B05G						
M2				B05H						
N2				B05J						
P2				B05K						
R2				B05L						
S2				B05M						
T2				A06U						
U2				A06V						
V2				A06W	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USACE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION E										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A1				A07X	N/A					
B1				A06X						
C1				A06Y						
D1				B06A						
E1				B06B						
F1				B06C						
H1				B06D						
J1				B06P						
K1				B06F						
L1				B06G						
M1				B06H						
N1				B06J						
P1				B06K						
R1				B06L						
S1				B06M						
T1				NONP						
U1				A06Z						
V1				A07F	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION E										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A2				NONE	N/A					
B2				A07Y						
C2				NONE						
D2				B05N						
E2				B05P						
F2				B05Q						
H2				B05R						
J2				B05S						
K2				B05T						
L2				B05U						
M2				B05V						
N2				B05W						
P2				B05X						
R2				B05Y						
S2				B05Z						
T2				A07Z						
U2				A07A						
V2				A07B	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4 FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
TSM TEST CONNECTOR 3-4										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
SECTION F										
A1				A07D	N/A					
B1				A07F						
C1				A07G						
D1				A05N						
E1				A05P						
F1				A05Q						
H1				A05R						
J1				A05S						
K1				A05T						
L1				A05U						
M1				A05V						
N1				A05W						
P1				A05X						
R1				A05Y						
S1				A05Z						
T1				NONE						
U1				A07J						
V1				A07L	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE / TEST CONN 3-4		USAGE		ZIF CONN. J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
TSM TEST CONNECTOR 3-4 SECTION F										CARD CAGE POSITION 3-4 IS A TSM CARD TEST CONNECTOR THIS CONNECTOR IS CONNECTED TO J17 AS SHOWN
A2				NONE	N/A					
B2				A07E						
C2				NONE						
D2				A05A						
E2				A05B						
F2				A05C						
H2				A05D						
J2				A05E						
K2				A05F						
L2				A05G						
M2				A05H						
N2				A05J						
P2				A05K						
R2				A05L						
S2				A05M						
T2				A07H						
U2				A07K						
V2				A07M	N/A					

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				A01J						
				A01K						
				A01L						
				B01M						
				B01W						
				B01X						
				B01Y						
				A02J						
				A02K						
				A02L						
				A02W						
				A02X						
				A02Y						
				A03S						
				A03U						
				A03V						
				A03W						
				A03X						
				A03Y						
				A03Z						
				B04E						
				B04F						
				B04M						
				B07A						
				B07B						
				B07C						
				A08A						
				A08B						
				A08C						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				A09A						
				A09B						
				A09N						
				A09P						
				A09Q						
				A09R						
				A09S						
				A09T						
				A09U						
				A09V						
				A09W						
				A09X						
				A09Y						
				A09Z						
				A10A						
				A10B						
				A10C						
				A10D						
				A10E						
				A10F						
				A10G						
				A10H						
				A10J						
				A10K						
				A10L						
				A10M						
				A10N						
				A10P						
				A10Q						
				A10R						
				A10S						
				A10T						
				A10U						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				A10V						
				A10W						
				A10X						
				A10Y						
				A10Z						
				B10A						
				B10B						
				B10C						
				B10D						
				B10E						
				B10F						
				B10G						
				B10H						
				B10J						
				B10K						
				B10L						
				B10M						
				B10N						
				B10P						
				B10Q						
				B10R						
				B10S						
				B10T						
				B10U						
				B10V						
				B10W						
				B10X						
				B10Y						
				B10Z						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				A11A						
				A11B						
				A11C						
				A11D						
				A11E						
				A11F						
				A11G						
				A11H						
				A11J						
				A11K						
				A11L						
				A11M						
				A11N						
				A11P						
				A11Q						
				A11R						
				A11S						
				A11T						
				A11U						
				A11V						
				A11W						
				A11X						
				A11Y						
				A11Z						
				B11A						
				B11B						
				B11C						
				B11D						
				B11E						
				B11F						
				B11G						
				B11H						
				B11J						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				B12G						
				B12H						
				B12J						
				B12K						
				B12L						
				B12M						
				B12N						
				B12P						
				B12Q						
				B12R						
				B12S						
				B12T						
				B12U						
				B12V						
				B12W						
				B12X						
				B12Y						
				B12Z						
				A13N						
				A13P						
				A13Q						
				A13R						
				A13S						
				A13T						
				A13U						
				A13V						
				A13W						
				A13X						
				A13Y						
				A13Z						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				B11K						
				B11L						
				B11M						
				B11N						
				B11P						
				B11Q						
				B11R						
				B11S						
				B11T						
				B11U						
				B11V						
				B11W						
				B11X						
				B11Y						
				B11Z						
				A12N						
				A12P						
				A12Q						
				A12R						
				A12S						
				A12T						
				A12U						
				A12V						
				A12W						
				A12X						
				A12Y						
				A12Z						
				B12A						
				B12B						
				B12C						
				B12D						
				B12E						
				B12F						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				B12G						
				B12H						
				B12J						
				B12K						
				B12L						
				B12M						
				B12N						
				B12P						
				B12Q						
				B12R						
				B12S						
				B12T						
				B12U						
				B12V						
				B12W						
				B12X						
				B12Y						
				B12Z						
				A11N						
				A11P						
				A11Q						
				A11R						
				A11S						
				A11T						
				A11U						
				A11V						
				A11W						
				A11X						
				A11Y						
				A11Z						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USAGE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				B13A						
				B13B						
				B13C						
				B13D						
				B13E						
				B13F						
				B13G						
				B13H						
				B13J						
				B13K						
				B13L						
				B13M						
				B13N						
				B13P						
				B13Q						
				B13R						
				B13S						
				B13T						
				B13U						
				B13V						
				B13W						
				B13X						
				B13Y						
				B13Z						
				B14C						
				B14D						
				B14E						
				B14F						
				B14G						
				B14H						
				B14J						
				B14K						
				B14L						
				B14M						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE		USAGE		ZIF CONN J17		RANGE				REMARKS
ZIF (J17)										
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				A15A						
				A15B						
				A15J						
				A15P						
				A15W						
				A15X						
				B15J						
				B15K						
				B15N						
				B15P						
				B15W						
				B15X						
				A16J						
				A16M						
				A16W						
				A16Z						
				B16J						
				B16M						
				B16W						
				B16Z						
				A17J						
				A17K						
				A17N						
				A17W						
				A17X						
				B17J						
				B17K						
				B17P						
				B17W						
				B17X						
				A18J						
				A18M						
				A18W						
				A18Z						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17)		USACE		ZIF CONN J17		RANGE				REMARKS
FUNCTION	CHANNEL	IN	OUT	SIGNAL	REF	IMPEDANCE	VOLTAGE	CURRENT	WAVEFORM	
SPARES				B18T						
				B18M						
				B18W						
				B18Z						
				A19T						
				A19K						
				A19W						
				A19X						
				B19T						
				B19K						
				B19W						
				B19Z						
				A20T						
				A20M						
				A20W						
				A20Z						
				B20T						
				B20M						
				B20W						
				B20Z						
				A22T						
				A22M						
				A22N						
				A22W						
				A22Z						
				B22M						
				B22W						
				B22Z						
				A23A						
				A23B						
				A23C						
				A23D						
				A23E						
				A23F						

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TABLE I. ZIF CONNECTOR SIGNAL INPUT/OUTPUT CHARACTERISTICS (cont)

TEST STATION MODULE ZIF (J17) FUNCTION	CHANNEL	USAGE IN	USAGE OUT	ZIF CONN J17 SIGNAL	ZIF CONN J17 REF	RANGE IMPEDANCE	RANGE VOLTAGE	RANGE CURRENT	RANGE WAVEFORM	REMARKS
SPARES				A23C						
				A23H						
				A23J						
				A23K						
				A23L						
				A23M						
				B23I						
				B23M						
				B23N						
				B23P						
				B23Q						
				B23R						
				B23S						
				B23T						
				B23U						
				B23V						
				B23W						
				B23X						
				B23Y						
				B23Z						

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TABLE II. INTERFACE DEVICE CHARACTERISTICS (TYPICAL)

Item	Manufacturer's Part Number	Functional Name
A	8T18	INTERFACE GATE
B	DM 8542	TRI-STATE QUAD I/O REGISTER
C	529A	HIGH SPEED VOLTAGE COMPARATOR
D	SN 74LS125	QUAD BUS BUFFER, TRI-STATE OUTPUTS
E	SN 7406	HEX DRIVER, OPEN COLLECTOR
F	SN 7414	SCHMITT TRIGGER INPUT GATE
G	9615PC or SN 75115	DIFFERENTIAL LINE RECEIVER
H	DM 8097	TRI-STATE HEX BUFFER
I	DM 8098	TRI-STATE HEX BUFFER
J	MM 74C902	CMOS-TO-TTL HEX INTERFACE
K	SN 74LS367	HEX BUS DRIVER, LOW POWER
L	SN 74LS368	HEX BUS DRIVER, LOW POWER
M	SN 74LS32	LOW POWER QUAD TWO INPUT OR GATE
N	SN 74LS04	LOW POWER, HEX INVERTER
O	DM 7542	QUAD TWO NAND, LOW POWER SCHOTTKY
P	SN 74LS01	LOW POWER, OPEN COLLECTOR QUAD TWO INPUT NAND GATE
Q	SN 74LS00	LOW POWER, QUAD TWO INPUT NAND GATE
R	LM339	QUAD COMPARATOR
S	9614PC or SN 75114	DIFFERENTIAL LINE DRIVER
T	DS 75461	DUAL PERIPHERAL DRIVER
U	DS 8831	TRI-STATE LINE DRIVER
V	74LS28	LOW POWER QUAD 2 NOR
W	SN 74LS14	LOW POWER, SCHMITT TRIGGER INPUT GATE

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TABLE III. DEFINITIONS

TTL LEVELS	
RECEIVER·	Logic "1" Input Voltage = +2.0 to +5.0V dc
	Logic "0" Input Voltage = 0.0 to +0.8V dc
DRIVER·	Logic "1" Output Voltage = +2.4 to +5.0V dc
	Logic "0" Output Voltage = 0.0 to +0.4V dc
STD TTL	
RECEIVER	Logic "1" Input Current = 40 UA, V(IN) = +2.4V dc
	Logic "0" Input Current = -1.6 MA, V(IN) = +0.4V dc
DRIVER	Logic "0" Output Current = 16 MA Min, V(OUT) = +0.4V dc
	Logic "1" Output Current = -400 UA Min, V(OUT) = +2.4V dc
LOW POWER SCHOTTKY	
RECEIVER	Logic "1" Input Voltage = +2.0 to +5.0V dc
	Logic "1" Input Current = 20 UA, V(IN) = +2.7V dc
	Logic "0" Input Voltage = 0.0 to 0.8V dc
	Logic "0" Input Current = -0.4 MA, V(IN) = 0.4V dc
DRIVER	Logic "1" Output Voltage = +2.4V dc Min, I(OUT) = -400 UA
	Logic "0" Output Voltage = +0.4V dc Max, I(OUT) = 8 MA
CMOS RECEIVER	
	Logic "1" Input Voltage = +3.5 to 5.0V dc
	Logic "0" Input Voltage = 0.0 to +1.5V dc
	Z(IN) = 10 MEG Min
INTERFACE GATE RECEIVER	
	Logic "1" Input Voltage = +9.0 to +30.0V dc
	Logic "1" Input Current = 50 UA Max
	Logic "0" Input Voltage = 0.0 to +6.0V dc
	Logic "0" Input Current = -1.8 MA Max, V(OUT) = +0.35V dc

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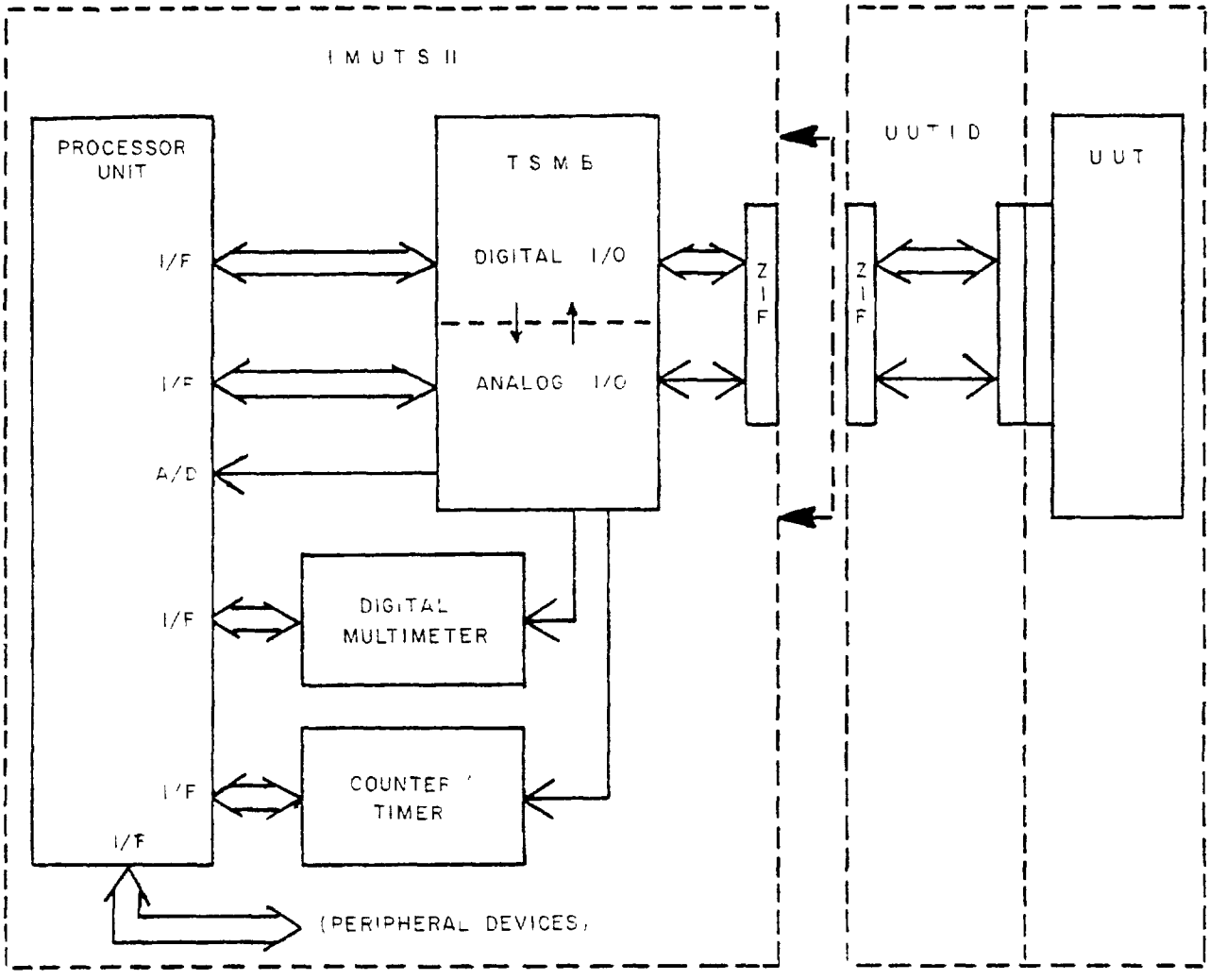


FIGURE 1 STANDARD TEST CONFIGURATION

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